

FUTURE REALM AGENCY

Platform Admin Panel

(Back Admin)

Master Business Requirements Document

The internal control centre for the Future Realm SMS ecosystem

FIELD	DETAIL
Document type	Platform Admin Panel (Back Admin) — Master BRD
Product	Future Realm School Management System
Version	4.0 — August 2026
Supersedes	Version 3.0 (2025). v3.0 is withdrawn and must not be built from.
Prepared by	Product Team — Future Realm Agency
Audience	Future Realm internal team only. Never visible to any school, partner or parent.
Status	Approved for build, subject to the Open Issues Register in Section 0.6
Classification	Confidential — Internal Use Only

WHY THIS VERSION EXISTS

The User Onboarding & Account Creation Specification v3.1 was written after this BRD and was approved for build. It reversed several decisions that v3.0 of this document still assumed — most importantly that a human approves every new school. Version 4.0 reconciles the two documents so that a developer reading both builds one product, not two.

It also closes fourteen structural gaps: capabilities that were promised in strategy, in the onboarding spec, or in the GSR counter-proposal, but which had no owning module here and would therefore simply never have been built.

Finally, it corrects the regulatory basis. Nigeria's data protection framework changed materially in September 2025 and v3.0 was written against the superseded instrument.

0 • Document Control

0.1 — Related documents and order of authority

Where two documents conflict, the higher document in this table wins. This ordering exists because the project has already produced one contradiction that would have reached the build.

RANK	DOCUMENT	STATUS	AUTHORITY
1	User Onboarding & Account Creation Spec v3.1	Final — approved for build	Absolute on identity, signup, provisioning, invitations, trial mechanics and login routing
2	Platform Admin Panel BRD v4.0 (this document)	Approved for build	Absolute on internal back-office operations
3	School Admin Portal BRD v1.0	Approved for build	Absolute on the school-facing portal
4	Foundation & Alignment Document v1.0	Requires update	Vision, pillars and market definition only. Its pricing and trial-length statements are superseded.
5	GSR Partnership Counter-Proposal	Issued	Commercial terms only, pending countersignature

DOCUMENTS REQUIRING UPDATE AS A CONSEQUENCE OF THIS VERSION

Foundation & Alignment Document v1.0 — §9 pricing framework (flagged, see 0.6), and the "one full academic term" trial statement, which is withdrawn.

School Admin Portal BRD v1.0 — three items: add a Never Activated list to Module 3 (staff) and Module 10 (guardians); extend Module 14.7 consent to all students under 18 rather than under 13; add the school-visible impersonation record described in Module 5.10 of this document.

0.2 — Changelog: what changed from v3.0 to v4.0

Every change below is a correction to a defect, a contradiction, or a gap. Nothing here is cosmetic.

A · Contradictions resolved against the Onboarding Specification

#	V3.0 SAID	V4.0 SAYS	WHY
A1	New school registrations enter a pending approval queue; no school is activated automatically (M2.2). Trial Active required Onboarding Specialist approval (M2.4).	Every school provisions instantly and unconditionally. There is no approval queue and no approval state. Risk assessment is automated, silent, and resolved during the trial before money changes hands (M2.9).	Onboarding Spec Rule R1. Manual approval does not scale, and there is no reliable Nigerian source against which a reviewer could actually verify a school. It was ceremony that delayed real customers.
A2	Trial = one full academic term, Starter features only (M3 subscription table).	Trial = 30 calendar days, every feature unlocked including SMS and WhatsApp, started only when the admin chooses to start it.	Onboarding Spec §9. A term-length trial creates a fairness problem with schools that pay immediately, and drags the decision across three months.
A3	Post-trial: read-only for 3 days, then locked (M3.5).	Post-trial: read-only for 7 days (to day 37), then locked. Data retained, never deleted. Reactivation restores everything.	Onboarding Spec §9 and Workflow 9.1.
A4	Account lifecycle ran Pending Verification → Trial Active → Active, with Rejected as a terminal state.	Lifecycle rebuilt (M2.5). Pending Verification and Rejected are removed as gating states. Provisioned, Trial Active, Dormant and Risk Flagged are added. Risk Flagged is an invisible attribute, not a state the school can perceive.	Aligns the state machine with instant provisioning.
A5	No dormancy concept existed.	Dormancy added (M2.12): 21 consecutive days without login during trial moves an account to Dormant; the web address is released 7 days after notice, and only if no real student records exist.	Onboarding Spec §8.3, plus a data-safety condition that spec did not state.

B · Modules and functions added because nothing owned them

#	ADDED	WHERE IT WAS PROMISED BUT HAD NO HOME
B1	M2.9 — Automated Risk Assessment & Review Queue	Onboarding Spec §8.2 specifies the queue and an internal SLA. No module owned it.
B2	M2.10 — Web Address Registry & Dispute Resolution	Onboarding Spec §3 and §8.3. The spec itself notes the dispute policy "must be written before launch".
B3	M2.11 — School Ownership Transfer	Nowhere. A school has one owner and one six-digit ID; proprietors sell schools, retire and die. No flow existed in any document.
B4	M2.12 — Dormancy & Address Reclamation	Onboarding Spec §8.3.
B5	M5.9 — Individual Users Register (self-registered teachers)	Named in the strategy as a ranked lead list sorted by school clustering. Absent from M5.
B6	M5.10 — School-Visible Support Access Record	Nowhere. v3.0 logged impersonation only to our own audit trail; the school was never told.

#	ADDED	WHERE IT WAS PROMISED BUT HAD NO HOME
B7	M13 — Partner, Channel & Commission Management (new module)	The GSR counter-proposal commits us to deal registration, attribution, clawback, commission on collected and reconciled revenue only, and a termly statement of account. None of it existed.
B8	M14 — Data Migration & Onboarding Operations (new module)	Strategy states free, effortless migration is a sales weapon. There was no migration tooling, pipeline or quality gate anywhere in the platform.
B9	M3.10 — Trial Pipeline Operations	Onboarding Spec §9 trial communication schedule had no operational owner.
B10	M11 data-scope dimension	The three-dimensional permission model is our stated moat. The internal panel used only two dimensions.

C · Regulatory corrections

#	CORRECTION	EFFECT ON THE BUILD
C1	v3.0 referenced NDPA 2023 and "NDPC requirements" generally. Nigeria's General Application and Implementation Directive (GAID) 2025 was issued 20 March 2025 and took effect 19 September 2025; on its issuance the NDPR 2019 and its 2020 Implementation Framework ceased to apply. NDPA 2023 and GAID together are now the complete framework.	M10 rebuilt around GAID's actual artefacts: registration, DPIA, DPO, breach notification, SNAG grievance handling, and cross-border transfer records.
C2	Registration as a Data Controller/Processor of Major Importance is triggered at more than 200 data subjects in a six-month window. The tiers are OHL (200–999), EHL (1,000+) and UHL (5,000+). Our client schools sit at OHL. Future Realm, aggregating students, guardians and staff across every school, reaches UHL almost immediately — the tier carrying the highest registration fees and an annual Compliance Audit Return.	NDPC registration and DPO appointment become launch requirements, not later commitments. Registration fees and DPO cost enter the financial model now. M10.6 tracks the filing calendar.
C3	GAID defines a child as anyone under 18 and requires parent or guardian consent to process a child's data regardless of the purpose of processing. The Onboarding Spec's under-13 consent trigger is the United States COPPA line and is wrong for Nigeria.	Guardian consent becomes universal for secondary students, not conditional on age. The age branch in Onboarding Workflow 5.4.3 collapses. School Admin BRD M14.7 must be widened correspondingly.
C4	M2.7 stated 12-month post-closure retention "per NDPC requirements". No such general retention period exists in the framework.	Restated as Future Realm retention policy with a documented lawful basis (M10.4). Inventing a regulatory source is precisely what an audit finds.
C5	M4.8 proposed selling partner access to a parent audience whose data the school controls and Future Realm merely processes.	Redesigned (M4.8). A processor cannot repurpose controller data for its own commercial benefit; a parent opt-in and a "Sponsored" label do not cure that. The school must opt in as controller and share revenue, or the feature does not ship.

D · Structural and format corrections

#	CORRECTION
D1	Module numbering 1–12 is deliberately unchanged so that every existing cross-reference from the Onboarding Spec and the School Admin Portal BRD still resolves. New modules are appended as 13 and 14.
D2	Every module now carries the full standing BRD structure, including the Offline Behaviour and Notification Triggers sections that v3.0 omitted entirely.

#	CORRECTION
D3	Data export corrected from "Excel and PDF" to Excel and CSV. PDF is a presentation format, not a data export format, and offering it as one breaks the promise that a school can take its data anywhere.
D4	Provider names (Termii, Brevo) removed from monitoring thresholds and moved to per-region configuration. The module monitors the configured provider for the region, whatever it is.
D5	Nigeria-specific verification signals, blocked-address lists and phone validation moved into a per-country configuration record (M9.7). None of them may be hard-coded.
D6	Pricing figures are marked provisional throughout and must not be built against until Open Issue 1 is closed.
D7	Module 9 rewritten and greatly expanded into a full curriculum configuration standard, including the Nigerian NERDC structure that took effect from the 2025/2026 session.

0.3 — How to read this document

Every module follows the same eight-part structure, so any developer, designer or team member opening any section knows where to find what they need.

SECTION	WHAT IT CONTAINS
Purpose	Why the module exists and which real-world problem it solves
Core Functions	Numbered, itemised capabilities. The numbering is stable and referenceable.
Workflows	State transitions and approval chains, drawn as diagrams
Key Data Entities	The database objects the module owns or consumes
Offline Behaviour	What works without a connection and what requires one
Notification Triggers	Which events fire messages, to whom, on which channel
Permissions Note	Which roles can do what, at what level, over which records
User Stories	Acceptance criteria written from the real user's perspective

A NOTE ON LANGUAGE

This document is read by developers, designers, school proprietors and commercial partners. It says "school account" rather than "tenant" and "web address" rather than "subdomain". Where a technical term is unavoidable it is explained at first use.

0.4 — Governing principles

These are decided. They are restated here because every module below is required to obey them, and because a specification that quietly reverses one of them is a defect.

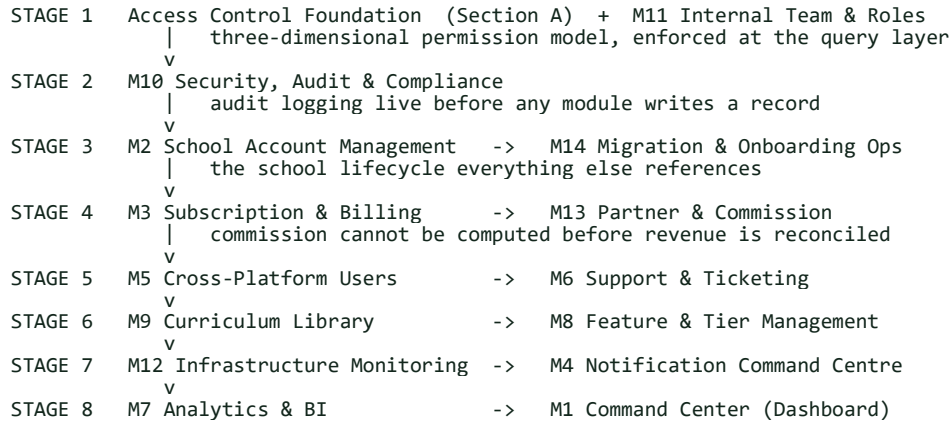
#	PRINCIPLE	WHAT IT FORBIDS
P1	Offline-first is the foundation, not a feature. Service worker caching, IndexedDB, a background sync queue that survives tab close, browser close and device restart, storage persistence requested on install, and sync status visible on every screen.	No school-facing workflow may assume connectivity. Attendance and score entry must work completely offline.
P2	Result computation correctness is existential. Deterministic engine, a stored calculation trace for every computed value, server-side as the sole authority, golden-dataset regression suite on every deployment. Rounding is half up, two decimals, at final total only.	No client-side computation is authoritative. No value may exist without a trace explaining how it was reached.
P3	Permissions are three-dimensional and enforced at the query layer: module access, function access and data scope, independently.	Hiding a button while the interface still returns the data is a security failure, not a permission system. This applies to the internal panel as much as to the school portal.
P4	Audit logging is written into every module from its first commit. Who, what, before, after, when, where and why. Immutable at the database layer.	No user, including a Super Admin and including the school owner, may alter an audit entry. Audit is never a later phase.
P5	Everything cultural is configuration, never code. Curriculum, grading, terms, calendars, currencies, languages, phone formats, date formats and report card layouts are all data.	Entering Ghana or Kenya must require a configuration record, not an engineering sprint. Any hard-coded assumption about Nigeria is a defect.

#	PRINCIPLE	WHAT IT FORBIDS
P6	The school owns its data absolutely. Full export at any time, without asking permission, in Excel and CSV.	No retention of a school's data as commercial leverage. No export gated behind a support ticket, a notice period or a settled invoice.
P7	Cost transparency before any paid action. Credit cost shown before every send, with balance before and after.	No school may discover it has spent money after the fact.
P8	Child safety and data protection are designed in. Where a feature would place Future Realm in the controller position for children's data with no school in the chain and no route to consent, it is redesigned architecturally, never defended legally.	Legal argument is not a substitute for a schema that cannot hold the data in the first place.
P9	Speed of the highest-frequency workflow determines adoption. Attendance is marked every day, in every class, often on a low-end Android phone.	Anything above sixty seconds per class for attendance is a failed design.

0.5 — Build sequence

Unchanged in principle from v3.0, with the two new modules placed where their dependencies allow. Access control is first because everything else is enforced through it, and retrofitting it is the single most expensive mistake available to this project.

PLATFORM ADMIN BUILD ORDER



M1 is built LAST on purpose: it is a read-only surface over every other module and cannot be finished before they exist.

Build sequence — access control and audit precede every module that writes data.

0.6 — Open Issues Register

These are carried forward unresolved. Each names an owner and the decision that is blocked behind it. No module below may be built in a way that assumes any of them is settled.

#	ISSUE	STATUS	BLOCKS
OI-1	Pricing. The per-student-per-term model produces annual costs to a school of roughly six to sixteen times observed Nigerian market rates. Correct modelling unit is annual cost per school — the only number the buyer ever sees. A base platform fee plus a smaller per-student component is under consideration.	Open. Verification of real rates through GSR outstanding.	M3 billing engine, M13 commission modelling, the revenue target, tier boundaries. The figures shown in M3.0 are placeholders and must not be built against.
OI-2	Free teacher tier data protection. With no school in the chain, Future Realm becomes controller for children's data with no route to guardian consent. Resolution is architectural: the tier must be structurally incapable of storing full names, photographs, dates of birth, admission numbers, guardian contacts or any sensitive data.	Design agreed, not yet specified in a BRD.	Teacher Portal BRD. Also constrains M5.9, which must never surface identifiable student data from individual accounts.
OI-3	NDPC registration tier and DPO appointment. Confirmed as launch requirements, not later commitments.	Action required.	Launch. M10.6 tracks it; someone must actually file.
OI-4	Web address dispute policy. The written procedure does not yet exist and has no owner.	Open.	M2.10. The first time a real school finds its name taken will arrive sooner than expected.
OI-5	The 50-real-student verification threshold is a chosen number, not a derived one.	Open — confirm with a data protection adviser.	M2.9 escalation trigger.

#	ISSUE	STATUS	BLOCKS
OI-6	Sample data is load-bearing for trial conversion, not a polish item.	Decided — recorded here as a build constraint.	It may not be descoped if the build runs late. Without it a 30-day trial never reaches the payoff moment.

SECTION A

Access Control Foundation

PURPOSE

Before any module exists, the platform must enforce who may enter it, what they may do inside it, and which records they may touch. These are three separate questions, and answering only the first two is how most systems leak.

This section is the specification the whole internal panel is enforced through. It is built first (Stage 1) because retrofitting a scope dimension onto a live permission system is the most expensive change available to this project.

CHANGED IN V4.0 *v3.0 specified two-dimensional access control — module and function. Data scope was missing, even though the three-dimensional model is the platform's stated competitive moat and is already specified for the school-facing portal. Running two different permission architectures across two products is a durable source of security bugs.*

A.1 — The three dimensions

DIMENSION	QUESTION IT ANSWERS	ENFORCED WHERE	EXAMPLE
Module access	May this person enter this module at all?	Navigation and route guard, then re-checked server-side on every request	A Support Agent cannot open Module 3 (Billing). The module is not in their navigation and the route rejects them.
Function access	Inside a module they can enter, what may they do?	Endpoint authorisation, per named function	A Support Agent inside Module 5 may force a password reset but may not suspend an account.
Data scope	Of the records this function can act on, which ones are theirs?	Injected into the database query itself — a WHERE clause the caller cannot influence	A Regional Growth Lead — Ghana runs the same school directory query as everyone else, but the query can only ever return Ghanaian schools.

THE ENFORCEMENT RULE THAT MATTERS

Scope is applied by the data layer, not by the caller. A permitted query is rewritten before execution to include the actor's scope predicate. There is no code path in which an application service can request records outside its scope, because the scope is not a parameter it supplies.

This is the difference between a permission system and a filtered interface. Competitors use a single role dropdown resolved in the front end. Retrofitting this later means rewriting every query in the system, which is why it is Stage 1.

A.1.1 — Available data scopes

SCOPE	MEANING	TYPICAL HOLDER
Global	Every record on the platform	Super Admin, CTO
Region	Schools within named countries, states or cities	Regional Growth Lead, country managers on expansion

SCOPE	MEANING	TYPICAL HOLDER
Portfolio	Only schools where this person is the assigned Account Manager	Customer Success, Account Managers
Assigned	Only records explicitly assigned to this person — tickets, migration jobs, review cases	Support Agents, Migration Specialists
Partner	Only schools attributed to one named partner under a registered deal	Partner-facing staff; the basis of the Phase 2 partner portal
Own	Only records this person created or acted on	Read-only advisors, contractors
None	No records; aggregate figures only, with no drill-through to an individual school	Analysts and advisors who need trends but not identities

SCOPE NONE IS NOT DECORATIVE

An advisor or contractor who needs to see churn trends does not need to see which school churned. Scope None returns aggregates with drill-through disabled and small-cohort suppression applied, so a chart with two schools in a segment cannot be used to identify them. This is the same discipline financial and health reporting use, and it is the correct default for anyone outside the core team.

A.2 — Internal role hierarchy

LEVEL	ROLE	CEILING	NOTES
1	Super Admin	Unrestricted — every module, every function, Global scope	The only role that can create internal accounts, approve data corrections, or force a school closure
2	Department Lead	Full within own department, at or below a ceiling set by Super Admin	May lower but never raise permissions within their team, and never above their own level
3	Team Member (Operational)	Function-scoped access within assigned modules	Scope is normally Assigned or Portfolio, not Global
4	Read-Only / Advisor	View-only dashboards and reports, zero action rights	Scope defaults to None. Granting Global scope to an advisor requires Super Admin action and a recorded reason.

Key rules

- Super Admin is the only role that can create a new internal account.
- A Department Lead can lower but never raise permissions within their team, and can never grant a permission or a scope they do not themselves hold.
- Custom roles can be created at any time as the team grows — for example Regional Growth Lead — Ghana, which is defined entirely by narrowing scope rather than by writing code.
- Every permission change, including every scope change, is written to Module 10 without exception and without a suppression option.
- There is no role that can grant itself a permission. Self-elevation is blocked at the data layer, not by convention.

A.3 — Default permission matrix

Defaults only. Every cell is individually adjustable per person by a Super Admin. Scope is shown because it is the dimension most often forgotten.

ROLE	SCOPE	FULL ACCESS	VIEW ONLY	NO ACCESS
Super Admin	Global	All modules	—	—
CTO / Product Lead	Global	M8, M9, M10, M12, M14	M1, M2, M7	M3, M13 amounts
Finance Lead	Global	M3, M13	M1, M2, M7 revenue	M6, M12, M5 actions
Customer Success Lead	Portfolio	M2 interventions, M6	M1, M3 status, M7 retention	M10, M12, M13
Support Lead	Assigned + escalation	M6, M5 account actions	M2 profile, M12 sync logs	M3 amounts, M10, M13
Support Agent	Assigned	M6 own tickets, M5 password reset	M2 overview, M5 registry	M3, M7, M10, M12, M13
Sales / Onboarding Lead	Region	M2 trial actions, M14, M13 deal registration	M3 trial status, M7 funnel	M10, M12, M3 pricing edits
Marketing Lead	Region	M4 campaigns	M7 engagement	M2 actions, M3, M5, M10
Data Protection Officer	Global	M10	M2, M5, M14	M3, M13

ROLE	SCOPE	FULL ACCESS	VIEW ONLY	NO ACCESS
Read-Only / Advisor	None	—	M1, M7 aggregates only	Everything else

ON THE DATA PROTECTION OFFICER ROLE

The DPO is a new internal role in v4.0 and is not optional. Under the current Nigerian framework a controller or processor of major importance must appoint one and register their details. Critically, the DPO must be able to see what is being processed without being able to change commercial terms — hence Global scope over Module 10 and read access elsewhere, with no billing rights at all. Independence is the point of the role; a DPO who reports into revenue is not a DPO.

MODULE 1

Platform Command Center

PURPOSE

The first screen every Future Realm team member sees on login. It answers one question instantly: is the business healthy right now? No navigation required — the pulse of the entire platform is visible from a single screen.

It owns no data of its own. Every figure on it is read from another module, which is why it is built last.

CHANGED IN V4.0 *Onboarding pipeline rewritten around instant provisioning rather than an approval queue. Risk review, migration health, dormancy, commission liability and compliance filing deadlines added. Every widget is now scope-filtered, not merely role-filtered.*

1.1 — Platform Pulse Strip

A persistent bar across the top, visible in every module, not only on the dashboard.

- Total active schools right now
- Total students across all active schools
- Schools currently online (active sessions in the last 30 minutes)
- System uptime, rolling 30 days
- Offline sync queue size — total records pending sync across all schools
- Last successful backup timestamp, with a red state if it is more than 6 hours old

1.2 — Revenue Snapshot

- Current month revenue collected, in Naira and in United States dollars
- Current term collected against total invoiced
- Total overdue balances
- Month-on-month revenue growth
- New recurring revenue added this month from conversions
- Notification credit revenue this month, shown separately from subscriptions
- Commission liability accrued this term (from M13) — *new in v4.0. Revenue that is collected but partly owed to a partner is not margin, and showing it as such flattens every figure above it.*

1.3 — Subscription Health

- Schools by tier as a donut: Starter, Standard, Elite, Trial, NGO
- Trials expiring within 7 days, each linking straight to the school profile
- Schools flagged as churn risk, from M3.6 scoring
- Schools in grace period — lapsed but still inside the read-only window
- Dormant accounts and addresses approaching release (M2.12)

1.4 — Onboarding & Activation Pipeline

Rewritten for instant provisioning. There is no queue of schools waiting for us; there is a set of schools we may be losing.

- Schools provisioned in the last 7 days
- Trials running, with day count and setup-wizard completion for each
- Schools stuck: no login for 5 or more days during trial, or setup wizard incomplete after 5 days
- Schools that have not yet started their trial — signed up but chose "Not yet"

- Risk-flagged accounts awaiting review, with days remaining before trial end (the internal SLA in M2.9)
- Migration jobs in progress and blocked (M14)
- Conversions this week

1.5 — Support Queue Snapshot

- Open tickets in total; critical tickets open; tickets breaching service level right now
- Average resolution time this week against last week

1.6 — System Health

- Interface response time and current error rate
- Sync failure rate over the last 24 hours
- Notification delivery success by channel
- Any active infrastructure alert, linking to M12

1.7 — Compliance Watch

New in v4.0. Regulatory deadlines are not a quarterly memory exercise.

- Days until the next NDPC registration renewal or Compliance Audit Return filing
- Open data subject requests and days remaining on each statutory clock
- Open grievances under the internal grievance mechanism, and their age
- Any unresolved breach notification obligation, shown in red and never dismissible

1.8 — Activity Feed

- Live scrolling log of significant platform events: school provisioned, payment received and reconciled, subscription lapsed, ticket opened, sync error detected, admin action taken, feature flag changed, migration completed, risk flag raised or cleared
- Every event timestamped and linked directly to its record
- The feed is scope-filtered: a Portfolio-scoped user sees events for their own schools only

1.9 — Geographic Distribution

- Map with school pins coloured by tier; clicking a pin opens the school profile
- City-level density view, because density is the growth strategy — fifty schools in one city are worth more than two hundred scattered across twelve states
- Map region is a configuration record, not a hard-coded Nigeria map

1.10 — Alert Centre

Everything requiring human action, in one place, each with a single-click action button.

ALERT	THRESHOLD	ROUTED TO
Trial expiring with no conversion signal	48 hours before expiry	Assigned Account Manager
Risk-flagged school not yet reviewed	7 days before trial end	Onboarding Lead — this is a hard internal SLA
Invoice overdue	Beyond 7 days	Finance Lead
Sync errors pending	More than 24 hours	CTO and Support Lead
Ticket unresolved beyond service level	On breach	Support Lead
School with zero logins	14 or more days	Customer Success
Migration job blocked	More than 48 hours	Migration Specialist and Onboarding Lead
Backup missed or failed	Immediately	CTO — critical, not clearable until a successful backup completes

ALERT	THRESHOLD	ROUTED TO
Data subject request approaching statutory deadline	7 days remaining	Data Protection Officer

Widget refresh rates

WIDGET	REFRESH
Revenue, subscription health	Every 15 minutes
Support queue, system health, activity feed	Real time, event driven
Churn risk flags	Daily recalculation
Geographic map	Hourly
Compliance watch	Daily, plus immediately on any status change
Commission liability	On each reconciliation event

Key Data Entities

This module owns no entities. It consumes from M2, M3, M6, M7, M10, M12, M13 and M14.

Offline Behaviour

None. The Platform Admin Panel is an online desktop tool used from the office on a reliable connection, and stating that plainly is deliberate: the offline-first principle governs the school-facing product, where teachers work on low-end phones in buildings with poor connectivity. There is no case for caching internal business data on staff devices, and doing so would widen our own data exposure for no operational gain.

Notification Triggers

EVENT	RECIPIENT	CHANNEL
Critical alert raised (backup failure, breach obligation, sync spike)	CTO, Super Admin	Email plus in-app, immediately
Risk review SLA at 7 days remaining	Onboarding Lead	Email plus in-app
Daily digest of alert centre items	Each Department Lead, scoped to their own alerts	Email, 07:30 local
Weekly business summary	Founders	Email, Monday 08:00

Permissions Note

- The dashboard is scope-filtered, not module-gated. Every internal user sees a dashboard; they do not all see the same one.
- A widget appears only if the user holds at least View access to the module that feeds it.
- Figures within a widget are further narrowed by data scope. A Portfolio-scoped Account Manager sees revenue for their own schools, not platform revenue.
- A Support Agent sees ticket and service-level widgets, and neither revenue nor infrastructure.
- Scope None users see aggregates with drill-through disabled.

User Stories

- As a Future Realm founder, I want to see revenue, active schools and commission liability the moment I log in, so that I know the true state of the business without treating partner-owned revenue as our own.
- As an onboarding lead, I want risk-flagged schools shown with days remaining before their trial ends, so that no account ever reaches payment without having been reviewed.

- As a product manager, I want to see schools stuck in setup, so that I can intervene personally before they abandon the trial.
- As a CTO, I want system health on the main dashboard, so that a sync or notification failure is caught by us before a school reports it.
- As a data protection officer, I want filing deadlines and open data subject requests on the dashboard, so that a statutory clock is never discovered after it has run out.
- As a regional lead, I want every widget narrowed to my region automatically, so that I am looking at my own business rather than filtering a national view every morning.

MODULE 2

School Account Management

PURPOSE

The master registry of every school on the platform, from the second they sign up to the day they leave. This module gives the team full lifecycle control, complete context on every school, and the ability to take any necessary action without a developer.

It is the module most changed by version 4.0, because the Onboarding Specification reversed its central assumption: nobody approves a school any more.

CHANGED IN V4.0 *The approval queue is removed. Schools provision instantly and unconditionally. Risk assessment is automated, invisible to the school, and resolved during the trial before any payment. Four new sub-modules are added: risk review (2.9), web address registry and disputes (2.10), ownership transfer (2.11) and dormancy (2.12).*

2.1 — School Directory

- Columns: School Name, City, State, Country, Tier, Status, Student Count, Setup Completion, Last Login, Renewal Date, Churn Risk, Risk Flag, Account Manager, Acquisition Source
- Filterable by tier, status, region, student count band, curriculum, date joined, acquisition source, partner, risk flag state, migration state
- Default sort: most recently active first
- Bulk select for mass notification, data export, or batch status action — every bulk action requires a typed confirmation naming the number of schools affected
- The directory query is scope-narrowed at the data layer. A Regional lead's directory cannot return a school outside their region, whatever they type into the filters.

2.2 — Provisioning Monitor

This replaces the v3.0 approval queue outright. There is no queue and no reviewer standing between a school and a working account. What the team watches instead is whether provisioning succeeded and whether the school is actually getting started.

The Onboarding Specification commits us to a complete, populated, working school in under sixty seconds with no human involvement. That is a promise this module must be able to prove we are keeping.

PROVISIONING STEP	OWNER	FAILURE HANDLING
School record created	System	Hard failure — signup cannot complete; user sees a retry, engineering is alerted
Web address reserved and made live with a security certificate	System	Hard failure — address reservation is atomic with school creation, never eventually consistent
Curriculum, grading and calendar templates loaded for the selected country	M9	Soft failure — school is usable; setup wizard prompts manual selection; alert raised
Default staff role templates installed	M11 templates, school copy	Soft failure — retried; blocks staff invitations until complete
Admin account created and identifier issued	System	Hard failure

PROVISIONING STEP	OWNER	FAILURE HANDLING
Sample data loaded	M14.6	Soft failure but treated as high severity — sample data carries trial conversion, so a school without it is at immediate risk and is flagged to the Account Manager
Trial record created if accepted	M3.10	Soft failure
Trial notification credit allowance issued	M3.7	Soft failure
Audit logging switched on	M10	Hard failure — a school account that is not being audited may not be handed to a user
Background risk assessment queued	M2.9	Soft failure, retried; never blocks the school

PROVISIONING HEALTH IS A MONITORED METRIC, NOT AN ASSUMPTION

M2.2 records elapsed time for every provisioning run and every step outcome. If the median exceeds forty-five seconds, or any step fails more than twice in an hour, an alert fires to the CTO. The sixty-second promise is made publicly in our onboarding material; it needs an instrument behind it.

2.3 — Full School Profile

Everything about one school on one screen, across seven tabs.

TAB	CONTENTS
Overview	School name, address, country, state, city, registration number where applicable, ministry approval number, proprietor name, primary phone, primary email, school type, curriculum, levels operated, date joined, tier, status, account manager, acquisition source and attributed partner
Configuration	Academic calendar and term dates, grading scale, assessment weighting, class and arm structure, subjects, report card template in use. Read-only to Future Realm staff — configuration belongs to the school. Where a school asks us to change it, the change is made by them with us guiding, or it goes through the data correction process in M6.6.
Usage	Module-level adoption, setup wizard completion, last result, fee and attendance activity, parent notification volume, tickets raised, login frequency over 30 days
Subscription	Tier, price, billing cycle, invoice and payment history, outstanding balance, trial dates, notification credit balances. Actions: extend trial, change tier, apply discount, record payment
Compliance	Consent coverage for the school's students, open data subject requests, the school's own data processing agreement version and acceptance date, retention clock if closing, export history
Migration	Source system, migration job status, records imported and rejected, reconciliation report. Empty for schools that started clean.
Activity Log	Complete timestamped audit trail — logins, publications, recordings, notifications, admin actions, support access sessions. Immutable and editable by nobody.

WHY THE CONFIGURATION TAB IS READ-ONLY

A school's grading scale, assessment weighting and term dates determine every result the platform computes. If a Future Realm employee can change them silently, then the correctness of a school's results depends on our internal discipline rather than on their own decision. Making the tab read-only is not a limitation; it is the mechanism by which the school, not us, owns the meaning of its own data.

2.4 — Account Status Management

The status set is rebuilt for instant provisioning. Every change carries a mandatory logged reason and is written to M10.

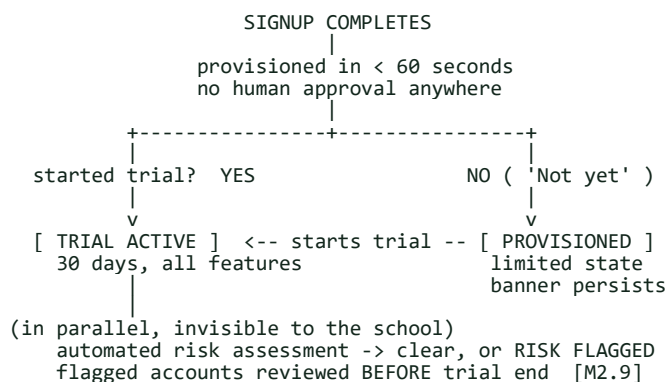
STATUS	MEANING	SET BY	SCHOOL'S EXPERIENCE
Provisioned	Account created, working, trial not yet started. The admin chose "Not yet".	System, at signup	Full account, limited feature state, persistent banner offering the trial
Trial Active	30-day trial running with every feature unlocked	System, when the admin starts the trial	Everything works
Active	Paying subscription in good standing	System, on payment reconciliation	Everything works
Grace Period	Subscription lapsed; read-only for 7 days	System, on overdue	Can see and export everything, cannot create
Trial Expired	30 days elapsed with no conversion; read-only for 7 days	System, at day 30	Can see and export everything
Dormant	No login for 21 consecutive days during trial	System (M2.12)	Account intact; web address at risk after notice
Suspended	Full access blocked	Finance Lead or Super Admin	Blocked, with export still available on request — data ownership is not conditional on payment
Reactivated	Restored from suspension, lock or dormancy	Finance Lead or Super Admin	Everything restored exactly as left
Locked	Day 37 after an unconverted trial, or day 7 after grace	System	No access; data retained, never deleted
Closed	School has formally left	Super Admin only, and only after export is delivered	Account gone; retention clock running per M10.4

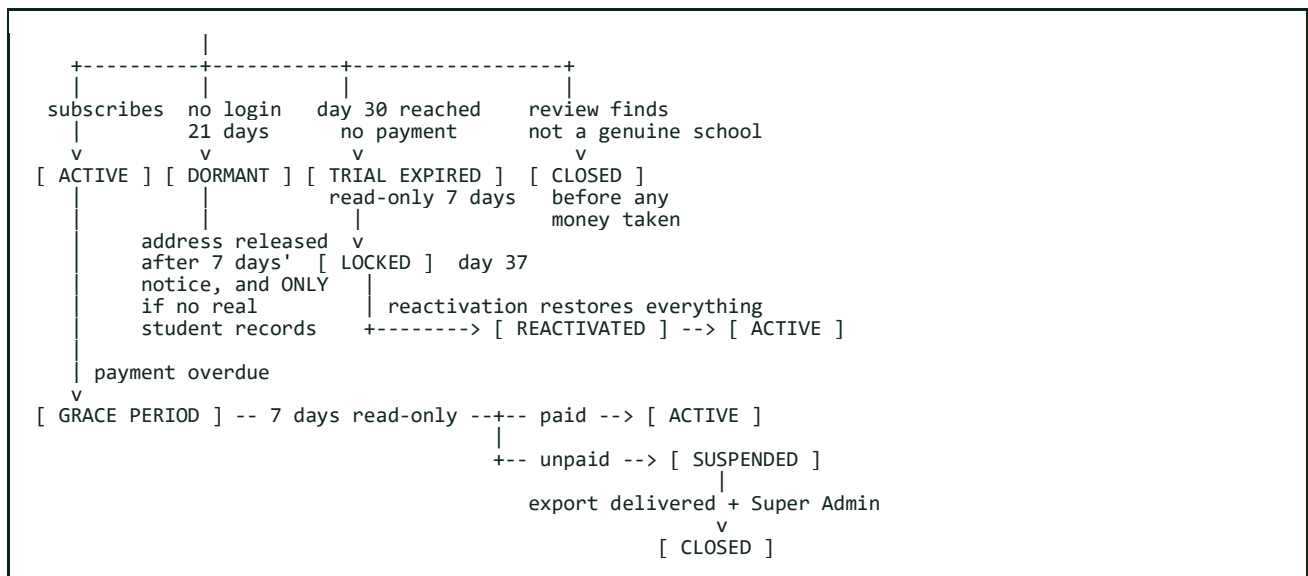
RISK FLAGGED IS AN ATTRIBUTE, NOT A STATUS

This is the single most important correction in this module. A flagged school appears in an internal queue and continues working completely normally — no restriction, no banner, no interruption, no altered behaviour of any kind. They never know. Treating suspicion as a visible state would punish the large majority of genuine schools that trip a signal for innocent reasons, and would do it at the exact moment we are trying to earn their trust.

2.5 — School Account Lifecycle

SCHOOL ACCOUNT LIFECYCLE — SIGNUP TO CLOSURE





Lifecycle v4.0 — Pending Verification and Rejected are gone. Risk assessment runs beside the lifecycle, never inside it.

2.6 — Manual Interventions

INTERVENTION	MINIMUM ROLE	CONSTRAINT
Extend trial	Sales / Onboarding Lead	Mandatory reason; extension count visible on the profile; a third extension requires Super Admin
Force tier upgrade or downgrade	Finance Lead	Mandatory reason; takes effect at next invoice unless overridden
Reset school admin password	Support Lead	Only via the verified identity process in M5.8
Apply promotional discount	Sales / Onboarding Lead	Must reference a campaign code from M3.9
Assign or change account manager	Customer Success Lead	Reassignment notifies both managers
Flag school for priority support	Support Lead	Raises service level; visible in M6 queue
Link school to a school group	Super Admin	Phase 2
Clear a risk flag	Onboarding Lead	Requires a recorded verification method and evidence note (M2.9)
Transfer school ownership	Super Admin	Full process in M2.11 — never a field edit

2.7 — Data Export and School Closure

- A full data export must be completed and delivered before any school can be closed. This is a gate in the workflow, not a courtesy.
- Export includes every student record, every result with its calculation trace, all fee history, all staff records, all attendance, all guardian links and the complete audit log for the school.
- Formats: Excel and CSV.** v3.0 said Excel and PDF. PDF is a presentation format; a school cannot load it into another system, so offering it as a data export quietly breaks the promise that the school owns its data.
- Export completion is logged and is a required precondition for a Super Admin to confirm closure. The system will not present the closure action until the export record exists.
- Export is available to the school itself at any time, from their own portal, without asking us and without a reason. This module is for the closure case; it is not the only route.
- Post-closure retention runs per M10.4 and is Future Realm retention policy under a documented lawful basis — not, as v3.0 stated, a general regulatory requirement.

2.8 — Multi-Branch and School Groups (Phase 2)

- A group allows several campuses under one billing account, each with its own data separation
- Group profile shows aggregate metrics across branches
- Billing consolidated at group level or managed per branch
- Designed for now, built later: the school record carries a nullable group identifier from the first commit, because adding a parent relationship to a live school table is far harder than carrying an unused column

2.9 — Automated Risk Assessment & Review Queue

NEW IN V4.0

The Onboarding Specification §8 defines automated risk checking and an internal review service level, and assigns them to no module. Without an owner this would not have been built, and the platform would have had instant signup with no fraud control at all.

Every school registers instantly and works normally. Assessment runs quietly in the background. Only flagged accounts are ever seen by a person, and only during the free trial — which is to say, before any money changes hands and therefore before there is anything to lose.

2.9.1 — Signals evaluated at signup

SIGNAL	METHOD	WEIGHT
Email domain matches the school's own website domain	String comparison	Strong positive
School website resolves and mentions the school name	Automated fetch	Strong positive
Business registration number supplied and correctly formatted	Format validation against the country's pattern	Positive
Ministry or education authority approval number supplied	Format validation	Positive
Phone number valid for the country and a recognised network prefix	Automatic	Positive
Generic free email, no website, no registration number	Automatic	Negative
School name closely matches an existing school in the same state	Fuzzy match	Flag — possible duplicate, handled by 2.9.3
Several signups from the same connection within 24 hours	Automatic	Negative
Same person registering a second or third school	Email, phone and device match	Flag
Blocked or protected web address requested	Blocklist (M2.10)	Blocked at signup — the only signal that stops anything

EVERY SIGNAL ABOVE IS COUNTRY-CONFIGURABLE

Registration numbers, ministry approval formats, phone prefixes and website conventions differ everywhere. The signal set, its thresholds and its validators are a configuration record per country (M9.7). Launching Ghana must mean adding a record, not editing a validator. This is the design constraint applied to the one place it is easiest to forget.

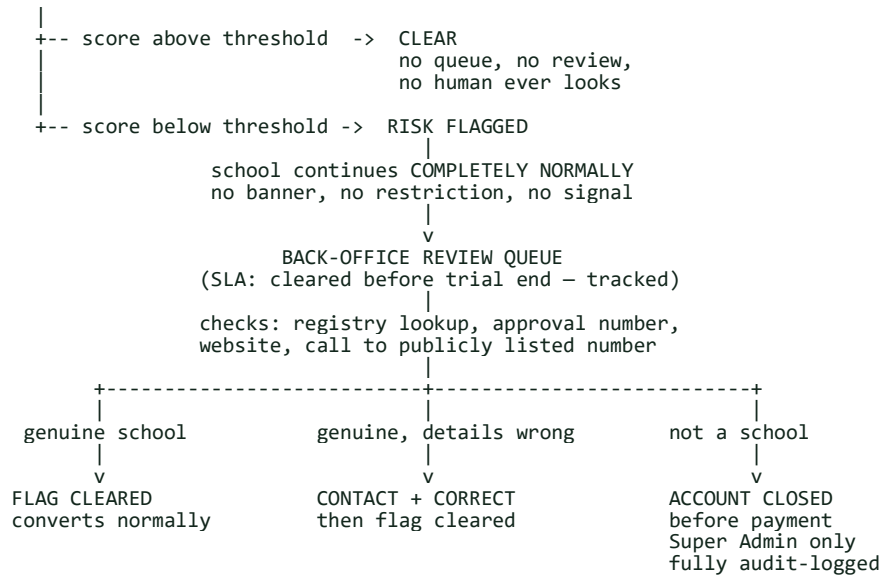
2.9.2 — Review workflow

RISK REVIEW — SILENT FOR THE SCHOOL, RESOLVED BEFORE ANY MONEY CHANGES HANDS

```

SIGNUP COMPLETE -> school is already provisioned and working
                    |      nothing in this workflow delays them
                    v
RISK ASSESSMENT RUNS (background, seconds)

```



Risk review — the school's experience is unchanged throughout, whatever the outcome.

2.9.3 — Automated protections requiring no human

RISK	AUTOMATIC CONTROL	WHY IT MATTERS
Two people from the same school both register	Fuzzy name and location match routes the second person to a join request against the first school instead of creating a duplicate	The highest-value control in the platform. Split school data — two half-populated accounts with the same name, neither correct — is an unrecoverable failure that destroys trust in everything else.
Someone claims an address belonging to a real school	Published dispute process (M2.10)	The first genuine school to find its name taken will arrive sooner than expected
Repeated trial abuse	Second and subsequent registrations from the same verified email, phone or device flagged automatically	Prevents indefinite free access through serial signup
Real children's data inside an unverified account	At 50 real student records, or at conversion to paid, verification becomes required	This is the point at which our data protection exposure genuinely begins. The threshold is a chosen number, not a derived one — see Open Issue 5.
Abandoned trial holding a good address	Dormancy (M2.12)	A squatted address is a loss to a real school later

2.9.4 — Escalation and the internal service level

- Review must be completed before the trial ends and before payment is taken. This is a tracked internal service level, surfaced on M1.10 at 7 days remaining.
- A flagged school that reaches 3 days before trial end without review escalates automatically to the Onboarding Lead and to Super Admin.
- A flagged school may not convert to paid while the flag stands. The block is applied in the billing module, not in the school's interface — the school sees a normal payment path and, in the rare case it is invoked, a support contact rather than an accusation.
- Closing an account under 2.9 is Super Admin only, requires the recorded evidence, and triggers immediate deletion scheduling under M10.4 rather than the standard retention clock, because there is no legitimate school whose records we would be preserving.

Where evidence is genuinely balanced, the current holder keeps the address and the claimant is offered a qualified alternative. Any other default invites a school to lose an address it has already printed on invitation emails, and makes us the arbiter of a commercial dispute we have no standing to judge. Reassignment happens only where the holder cannot evidence a legitimate claim at all.

2.11 — School Ownership Transfer

NEW IN V4.0 — THIS GAP EXISTED IN EVERY DOCUMENT

A school account has one owner and one six-digit identifier. Proprietors sell schools, retire, fall ill and die. In the Nigerian private school market this is not an edge case; it is a normal event in the life of a business. No document specified what happens, which means today the answer would be an undocumented database edit by whoever was available — the worst possible answer for a record that determines who controls a school's children's data.

Ownership transfer is never a field edit. It is a controlled process with evidence, dual approval and a notification obligation, because the owner is the person who holds the school's data controller responsibility.

TRIGGER	EVIDENCE REQUIRED	APPROVAL
Voluntary sale or handover	Signed transfer instruction from the current owner, plus identity confirmation of both parties	Super Admin, after Onboarding Lead verification
Owner incapacitated	Board resolution or legal instrument naming the successor, plus school registration document	Super Admin plus a second Super Admin or the CTO — dual approval
Owner deceased	Letters of administration, probate, or an equivalent instrument; plus registration document	Dual approval, and a mandatory 14-day hold before the change takes effect
Dispute between claimants	No transfer is made. The account is frozen in read-only, both parties notified, and the matter awaits a documented resolution between them.	Super Admin freeze only

OWNERSHIP TRANSFER

```

REQUEST RECEIVED (any of the four triggers above)
  |
  v
EVIDENCE COLLECTED AND RECORDED    <---- incomplete: request held,
  |                                   outgoing owner notified
  v
IDENTITY VERIFIED - both outgoing and incoming owner
  |                   incoming owner verified by the M5.8 process
  v
NOTICE TO THE OUTGOING OWNER'S REGISTERED CONTACTS
14 days where the owner is deceased or incapacitated
immediate where the transfer is voluntary and signed
  |
  |-- objection raised --> ACCOUNT FROZEN, read-only,
  |                       no transfer until resolved
  |
  v
DUAL APPROVAL (Super Admin + second approver)
  |
  v
TRANSFER EXECUTED
- new owner account created and linked; NOT a reassigned login
- outgoing owner's access revoked, sessions terminated
- school identifier, web address and all data unchanged
- data processing agreement re-accepted by the new owner
- all school staff notified that ownership has changed
  |
  v
WRITTEN TO AUDIT LOG - evidence references, both approvers,
notice dates, objection record if any. Permanently retained.

```

Ownership transfer — evidence, notice, dual approval, and a new account rather than a handed-over login.

WHY THE INCOMING OWNER GETS A NEW ACCOUNT, NEVER THE OLD LOGIN

Handing an existing login to a different human being destroys the audit trail: every action taken before the transfer becomes attributable to the wrong person, and every action after it is indistinguishable from the previous

owner's. Since the audit trail is what protects the school in a dispute, and protects us in a regulatory inspection, the login must change even though it is the slower option.

2.12 — Dormancy & Address Reclamation

NEW IN V4.0

Specified in Onboarding §8.3 with no owning module — and specified without the data-safety condition that this section adds.

STAGE	TRIGGER	EFFECT
Inactivity watch	10 consecutive days with no login during trial	Internal only. Account Manager alerted; no message to the school beyond the normal trial communication schedule.
Dormant	21 consecutive days with no login during trial	Status set to Dormant. Account remains fully intact and instantly restorable. Email and message sent giving 7 days' notice that the web address will be released.
Address released	7 days after notice, and only if the conditions below are all met	Address returns to the available pool. School data is retained in full and is unaffected.
Restored	Any login during or after dormancy	Status returns immediately to its previous value. If the address was already released, a new address is issued and the school is told plainly why.

THE CONDITION THE ONBOARDING SPECIFICATION DID NOT STATE

An address is released only where the school holds no real student records, no real staff accounts beyond the founding admin, and has sent no invitations to third parties. Releasing an address behind which real children's data sits would mean a stranger could later take a name that a real school's parents already associate with their children — and would strand every invitation link already in circulation. Where real data exists, the address is held indefinitely and the case is escalated to the Onboarding Lead as a human decision instead.

Key Data Entities

ENTITY	KEY FIELDS
School	school_id, name, registration_no, ministry_approval_no, address, city, state, country_code, curriculum_template_id, levels_operated[], enrolment_count, tier, status, risk_flag_state, setup_completion_pct, churn_risk_score, created_at, account_manager_id, acquisition_source, partner_id (nullable), parent_group_id (nullable)
School Contact	contact_id, school_id, name, role, phone, email, is_primary, verified_at
Provisioning Record	school_id, started_at, completed_at, elapsed_ms, step_results (json), failures[], retried_at
Risk Assessment	assessment_id, school_id, score, signals (json), threshold_applied, flagged (bool), assessed_at
Risk Review	review_id, school_id, reviewer_id, methods_used[], evidence_notes, outcome, resolved_at, sla_met (bool)
Web Address	address, school_id (nullable), state (live/held/retired/reserved/blocked), issued_at, retired_at, release_eligible_at, redirect_from, redirect_expires_at, country_scope
Address Dispute	dispute_id, claimed_address, claimant_details, evidence_refs[], holder_notified_at, holder_response, outcome, decided_by, decided_at

ENTITY	KEY FIELDS
Ownership Transfer	transfer_id, school_id, outgoing_owner_id, incoming_owner_id, trigger_type, evidence_refs[], notice_sent_at, hold_expires_at, objection (nullable), approver_1_id, approver_2_id, executed_at
Dormancy Record	school_id, dormant_since, notice_sent_at, address_release_due, real_data_present (bool), released_at (nullable), restored_at (nullable)
Account Activity Log	log_id, school_id, actor_id, action_type, before_state, after_state, reason, details, timestamp

Offline Behaviour

None — internal desktop tool. Note the asymmetry deliberately: the school-side experience of every state in 2.4 must work offline. A school in Grace Period or Trial Expired must still be able to open and export its data on a poor connection, because read-only access that requires connectivity is not access.

Notification Triggers

EVENT	RECIPIENT	CHANNEL	TIER
School provisioned	School admin	Email plus WhatsApp or SMS	All
Provisioning step failed	CTO, Account Manager	In-app plus email	Internal
Trial started	School admin	Email	All
Trial day 21, 27, 30, 37	School admin	Per the Onboarding trial schedule	All
Dormancy notice	School admin	Email plus WhatsApp or SMS	All
Address release	School admin	Email plus WhatsApp or SMS	All
Status changed by an admin	School admin	Email, always, with the reason where it is appropriate to share	All
Ownership transfer executed	All school staff	Email plus in-app	All
Account manager reassigned	Both managers	In-app	Internal
Export ready before closure	School admin	Email with a time-limited secure link	All

Permissions Note

- View directory and profile overview and usage tabs: all internal roles, narrowed by data scope
- View subscription tab: Finance Lead and above
- View compliance tab: Data Protection Officer, Super Admin, and the assigned Account Manager for their own schools
- Change status (suspend, lock, reactivate): Finance Lead or Super Admin
- Extend trial or apply discount: Sales / Onboarding Lead and above
- Clear a risk flag: Onboarding Lead and above, with recorded evidence
- Execute an ownership transfer: Super Admin, with a second approver
- Close a school: Super Admin only, and only after a delivered export
- Nobody at any level may edit the Configuration tab. There is no permission that grants it.

User Stories

- As an onboarding lead, I want risk-flagged schools listed with days remaining before their trial ends, so that every account is reviewed before it becomes a paying customer, without a single genuine school being delayed at signup.
- As a customer success lead, I want one school's full profile on one screen, so that I am always informed and prepared before a call.

- As a finance lead, I want to suspend access when a school is significantly overdue, so that we protect revenue without engineering involvement — and the school can still export its own data.
- As a sales lead, I want to extend a trial by two weeks as a retention gesture, so that I can close a difficult renewal without a developer.
- As a support agent, I want a school's last login date and module usage on their profile, so that I can diagnose low adoption before it becomes churn.
- As a Super Admin, I want ownership transfer to require evidence, notice and two approvers, so that a school changing hands never turns into a database edit nobody can account for.
- As an onboarding specialist, I want the second person from the same school to be offered a join request rather than a new account, so that a school's data is never split across two half-populated accounts.
- As a proprietor whose school has left the platform, I want a complete export in Excel and CSV before my account closes, so that my school's records are mine and I can take them anywhere.

MODULE 3

Subscription & Billing Management

PURPOSE

Governs the entire commercial relationship between Future Realm and every school: what they pay, when they pay, what happens when they do not, and what the business is earning in aggregate.

Because billing is per student per term and student counts change every term, this module must be both precise and flexible. Because commission is paid to partners out of it, it must also be able to distinguish money invoiced from money actually received.

CHANGED IN V4.0 Trial corrected to 30 days, all features, admin-started, with a 7-day read-only tail. Pricing marked provisional pending Open Issue 1. Reconciliation elevated from a status to a distinct, evidenced event because partner commission depends on it. Trial pipeline operations added as 3.10.

3.0 — Subscription structure**THESE FIGURES ARE PROVISIONAL AND MUST NOT BE BUILT AGAINST**

Modelled annually, the current per-student-per-term figures produce a cost to a school of roughly six to sixteen times observed Nigerian market rates. The buyer — a proprietor or principal — never sees a per-student figure. They see one annual number, and that is the number the model must be built on.

The billing engine must therefore be built to be indifferent to the pricing shape: a base platform fee, a per-student component, a band price, or any combination, all expressed as data. Hard-coding the structure below into the engine would mean rebuilding it the moment Open Issue 1 closes.

TIER	STUDENT RANGE	PROVISIONAL PER STUDENT PER TERM	ADD-ONS UNLOCKED
Starter	1 – 250	Provisional — see Open Issue 1	Email notifications, all core modules
Standard	251 – 700	Provisional	Adds SMS and WhatsApp notifications, payment integration
Elite	701 +	Provisional	Adds predictive analytics, custom branding, dedicated account manager
NGO / Mission	Any size	90% off the applicable tier	Verified annually — registration or charity certification required. Not free for life; verification lapses and the discount lapses with it.
Trial	Any size	Free	30 days, every feature unlocked, including SMS and WhatsApp against a fixed credit allowance

Additional billing rules

- Annual prepay discount where all terms are paid in advance
- Volume relief as student count rises within a tier, expressed as a price band table rather than a formula in code
- Notification credits for SMS and WhatsApp are purchased separately as usage-based bundles and are never bundled into the subscription price

- Student count is confirmed at the start of each term. The system recommends a tier change; it never applies one automatically. A school discovering an automatic upgrade on an invoice is a support ticket and a trust loss.
- Every price is held in the school's own currency alongside a reference currency. Currency is a configuration record, not a display formatting choice.

3.1 — Subscription Dashboard

- Monthly and annualised recurring revenue, in local currency and reference currency
- Revenue split by tier, and separately by school size band — these are not the same question
- Average revenue per school
- Notification credit revenue this month
- Churn rate last term, as a percentage of schools that did not renew
- Projected renewal revenue next term
- Commission liability accrued and commission settled, from M13
- Net revenue after commission — shown alongside gross, never instead of it

3.2 — Invoice Management

Invoices are generated each term for schools renewing or converting from trial, based on confirmed student count and current tier. No invoice reaches a school without an admin reviewing it.

STATUS	MEANING
Draft	Generated, not yet reviewed
Sent	Delivered to the school admin
Pending Payment	Sent, awaiting payment
Partially Paid	Payment received below the total. No commission accrues on a partial payment.
Paid	Full payment received but not yet matched to a bank record
Reconciled	Payment confirmed and matched to a bank record. This is the only status that recognises revenue and the only one that accrues commission.
Overdue	Not received by the due date
Escalated	Overdue beyond grace; account action required
Cancelled	Voided, with a logged reason and an immutable record of the original

PAID AND RECONCILED ARE DELIBERATELY DIFFERENT STATES

An invoice marked Paid records that someone told us money arrived. Reconciled records that we matched it to a bank statement line. Partner commission, revenue recognition and every figure presented to an investor derive from Reconciled and never from Paid.

This is the same distinction the GSR counter-proposal turns on — commission is payable on collected and reconciled revenue, never on invoiced revenue. Collapsing the two states here would quietly break the commercial agreement in the software.

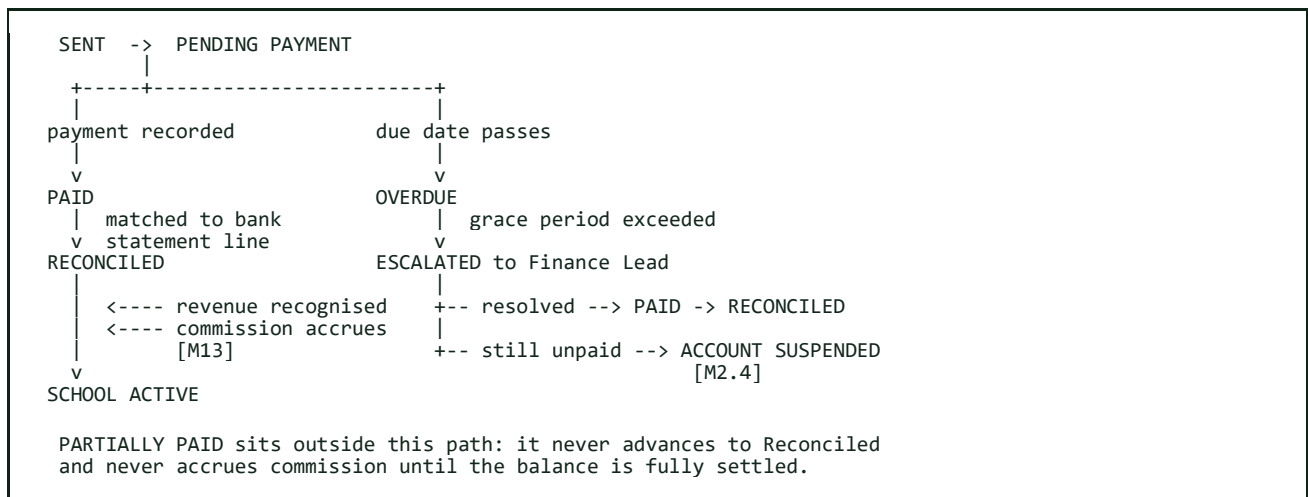
3.3 — Invoice lifecycle

INVOICE LIFECYCLE — DRAFT TO RECONCILIATION OR ESCALATION

```

INVOICE GENERATED (term rollover, confirmed student count, current tier)
  |
  v
DRAFT --- admin reviews; no invoice leaves the platform unreviewed
  |
  v

```



Invoice lifecycle — reconciliation, not payment, is the event that means something.

3.4 — Manual Payment Recording and Reconciliation

Automated payment integration is deferred to Phase 2, so payments are recorded by hand. That makes the controls around it more important, not less.

- Admin opens the invoice and enters payment date, amount, method, bank reference and their own identity
- Subscription status updates immediately and a payment confirmation goes to the school admin
- Reconciliation is a separate action, performed by a different person from the one who recorded the payment. Segregation of duties is not bureaucracy here: this is the only control standing between a manual payment field and a fabricated revenue figure.
- Reconciliation requires a bank reference and records the matcher's identity and timestamp
- Any payment unreconciled after 7 days appears on the Finance Lead's exception list
- An amended or reversed payment never overwrites the original — it writes a correcting entry, and both remain visible

3.5 — Trial Management

ITEM	SPECIFICATION
Length	30 calendar days
Starts	Only when the admin chooses to start it. Never automatically on signup.
Features	Every module unlocked, including SMS and WhatsApp notifications
Credits	A fixed trial allowance sized for genuine testing, not bulk sending
Payment details	Not required to start
Alerts	Fire at 7 days and 2 days before expiry, to the assigned Account Manager and the M1 alert centre
At day 30	Read-only for 7 days. Nothing is lost and everything remains visible and exportable.
At day 37	Locked. Data retained per policy and never deleted. Reactivation restores everything.
Extension	Available to Sales / Onboarding Lead with a recorded reason. A third extension requires Super Admin.
Blocked conversion	A school with an open risk flag cannot convert to paid until the flag is cleared (M2.9.4)

3.6 — Churn Risk Scoring

Recalculated daily for every school from behavioural signals.

SIGNAL	WEIGHT	APPLIES
No admin or teacher login for 10 or more days	High	Any active-term school
No result or score entry for 14 or more days mid-term	High	During active term only
No fee recording activity for 14 or more days	Medium	During active term only
Trial nearing expiry with low engagement	High	From 7 days before trial end
Setup wizard still incomplete after 14 days	High	New in v4.0. A school that never finished setup never reached the product's value.
Support ticket unresolved for 5 or more days	Medium	Any open critical ticket
No parent notification sent all term	Low	After week 4 of term
Migration job blocked or abandoned	High	New in v4.0. A displaced school whose data never arrived has not actually joined.

- High scores are flagged on M1, in the school directory, and in a weekly digest to Finance and Customer Success
- The score resets at the start of each term
- The score is never shown to the school and never used to alter their experience — it drives our outreach, not their product

3.7 — Notification Credit Wallet Management

- Every school's SMS and WhatsApp balance visible from their profile and from this module
- Manual top-up after confirming a bundle payment, following the same record-then-reconcile discipline as 3.4
- Low-balance alert fires below 50 messages, to the school and to their Account Manager
- Consumption history shows what consumed credits — result alerts, fee reminders, announcements — so a school can see where its money went
- Rate card, showing what Future Realm pays the provider against what the school pays, visible to Super Admin and Finance Lead only
- **Margin alert:** if a provider's cost per message rises above a configured percentage of the school price, an alert fires to Finance and the CTO. Provider pricing moves; discovering a negative margin at quarter end is avoidable with one threshold.

3.8 — Revenue Reporting

- Revenue by tier, by size band, by state and city, by acquisition source, and by partner
- Gross revenue, commission liability, and net revenue after commission — all three, always together
- Total outstanding receivables and ageing
- Term-over-term renewal rate
- Lifetime value projection by tier, based on observed subscription duration rather than assumption
- All reports exportable to Excel and CSV

3.9 — Discount and Promotion Management

- Super Admin creates codes with discount type, value, expiry and maximum redemptions
- Every redemption logged with school, applying admin and campaign
- Campaign reporting shows redemptions, total discount issued, and schools converted
- A discounted invoice records both the list price and the discounted price, because commission and revenue reporting need the difference

3.10 — Trial Pipeline Operations

NEW IN V4.0

The Onboarding Specification defines a trial communication schedule spanning day 0 to day 37 and assigns it no operational owner. Left unowned, the schedule becomes a marketing intention rather than a system behaviour.

DAY	MESSAGE	CHANNEL	FIRES FROM
0	Welcome, school web address, identifier, what to do first	Email plus WhatsApp	M2 provisioning
1	Download your first report card — guided prompt against sample data	Email plus in-app	M3.10
3	Import your students — migration guide	Email plus in-app	M3.10, links to M14
7	Invite your teachers	Email plus in-app	M3.10
14	Progress summary — what the school has achieved so far	Email	M3.10
21	Nine days left, plus pricing for their student count	Email plus WhatsApp	M3.10
27	Three days left, plus payment instructions	Email plus WhatsApp	M3.10
30	Trial ended — read-only, seven days to reactivate	Email plus WhatsApp	M3.5
37	Account locked, data safely retained	Email	M3.5

- Every message in the schedule is a template in M4, not text embedded in code, so it can be edited without a deployment
- Messages are suppressed where they would be absurd — the day 3 import prompt does not fire at a school that has already imported 400 students
- The pipeline view shows every trial school against the schedule, with sends completed, upcoming and failed
- A school that has not started its trial receives none of this. The Provisioned state has its own short sequence encouraging them to begin.

Key Data Entities

ENTITY	KEY FIELDS
Subscription	school_id, tier, pricing_model_id, price_components (json), currency, discount_type, billing_cycle, start_date, end_date
Pricing Model	model_id, name, country_code, currency, base_fee, per_student_bands (json), effective_from, effective_to, created_by
Invoice	invoice_id, school_id, term, student_count, list_amount, discount_amount, amount_due, currency, status, due_date, issued_at, issued_by, partner_id
Payment	payment_id, invoice_id, amount, method, bank_reference, recorded_by, recorded_at, reconciled_by, reconciled_at, correcting_entry_for (nullable)
Trial	school_id, offered_at, started_at, ends_at, converted (bool), conversion_date, extensions[] (extended_by, reason, new_end_date)
Churn Risk Score	school_id, score, signals (json), last_calculated
Notification Wallet	school_id, sms_balance, whatsapp_balance, low_balance_threshold, last_topped_up, provider_cost_snapshot
Promo Code	code, discount_type, value, campaign_name, valid_from, valid_to, max_redemptions, redemption_count

ENTITY	KEY FIELDS
Trial Pipeline Event	school_id, day_index, template_id, scheduled_for, sent_at, suppressed_reason (nullable), delivery_status

Offline Behaviour

None internally. On the school side, an unpaid or expired school must retain offline read and export access to its own data — the commercial state of an account never removes a school's ability to reach its records.

Notification Triggers

EVENT	RECIPIENT	CHANNEL
Invoice sent	School admin and billing contact	Email plus WhatsApp
Payment recorded	School admin	Email
Payment reconciled	School admin, plus partner if attributed	Email
Invoice overdue at 3 and 7 days	School admin, then Finance Lead	Email
Trial expiry at 7 and 2 days	Account Manager and school admin	In-app and email
Credit balance below threshold	School admin and Account Manager	In-app plus email
Provider margin threshold breached	Finance Lead and CTO	Email, immediate
Payment unreconciled after 7 days	Finance Lead	Email digest

Permissions Note

- View and edit pricing, issue refunds: Finance Lead and Super Admin only
- Record a manual payment: Operations Admin and above
- Reconcile a payment: Finance Lead and above, and never the same person who recorded it — enforced by the system, not by policy
- View trial status and apply promotion codes: Sales / Onboarding Lead and above
- View churn risk scores: Customer Success and above
- View the provider rate card: Super Admin and Finance Lead only
- Portfolio-scoped users see billing for their own schools only, including on every aggregate figure

User Stories

- As a finance lead, I want every school's invoice status in one view, so that I know exactly who is paid, pending or overdue without chasing schools individually.
- As a finance lead, I want reconciliation to be a separate action from recording a payment, performed by a different person, so that our revenue figures are evidenced rather than asserted.
- As a customer success manager, I want an automatic alert when a trial is seven days from expiring, so that I can reach out before a school churns silently.
- As a retention lead, I want schools with dropping engagement flagged before their renewal date, so that I can intervene while there is still a relationship to save.
- As an operations lead, I want to top up a school's WhatsApp balance after confirming payment, so that parent notifications are never interrupted by a billing delay.
- As a Super Admin, I want gross revenue, commission liability and net revenue shown together, so that I never present a partner's share of revenue as our margin.
- As a proprietor, I want to see exactly which messages consumed my credits, so that I can see where my money went rather than trusting a balance that only ever falls.

MODULE 4

Notification & Communication Command Center

PURPOSE

Future Realm's own channel into every school on the platform: operational alerts, product announcements, onboarding guidance and renewal reminders.

Version 3.0 also proposed evolving this into a monetised advertising channel reaching parents. That proposal is redesigned in 4.8 for reasons that are not negotiable.

CHANGED IN V4.0 *Commercial campaigns to parents redesigned. Message templates unified with the trial pipeline. Provider names removed from the specification. Consent handling restated around the correct controller and processor relationship.*

4.1 — Two types of communication

TYPE	PURPOSE	EXAMPLES	APPROVAL
Operational	Platform management and school success	Maintenance notice, renewal reminder, feature release, onboarding guidance	Department Lead
Promotional	Future Realm's own products and offers, to school staff only	Tier upgrade offer, training webinar, new module launch	Super Admin or Marketing Lead
Commercial (third party)	Revenue-generating partner messages	See 4.8 — not permitted to parents in the form v3.0 proposed	Super Admin, plus the school as controller

- Promotional communications are opt-in only and opt-out is honoured immediately and permanently
- Opt-out exclusions are enforced at system level. There is no permission that overrides them for any role.
- Operational messages — maintenance, billing, security — bypass promotional opt-out, because they concern the service the school is paying for. This distinction is written into the template itself, not decided at send time.

4.2 — Audience Builder

- Filter by user role, tier, country, state, city, individual school, student count band, last login range, renewal date range, engagement score range, migration state, setup completion, and promotional opt-in status
- Live recipient count shown before any send, updating as filters change
- Estimated credit cost shown before any send, with the balance before and after — the same cost transparency principle we require of schools applies to us
- Audiences are saved as reusable segments with their filter criteria, not as frozen recipient lists that silently go stale

4.3 — Message Composer

CHANNEL	FORMAT	LIMITS	RULES
Email	Rich text with subject and optional attachment	Use judgement	Must carry an unsubscribe link for anything promotional
SMS	Plain text	Multi-part warning shown above single-message length	No images; links only as shortened URLs

CHANNEL	FORMAT	LIMITS	RULES
WhatsApp	Approved templates only	Per template approval	Every template pre-approved by the provider before use
In-app	Rich text banner or modal	One active banner per surface	Dismissible unless it is a compliance or security notice

- Live preview showing exactly how the message renders on the recipient's device, at the smallest supported screen size first
- Variable fields — school name, amount due, term dates — auto-populated from the school record, with a preview against a real record before send
- Sender name is always Future Realm. Individual staff names are never used, so that no message can be impersonated by someone claiming to be a named employee.
- Every message body is language-tagged and translatable. Message text lives in template records, never in code.

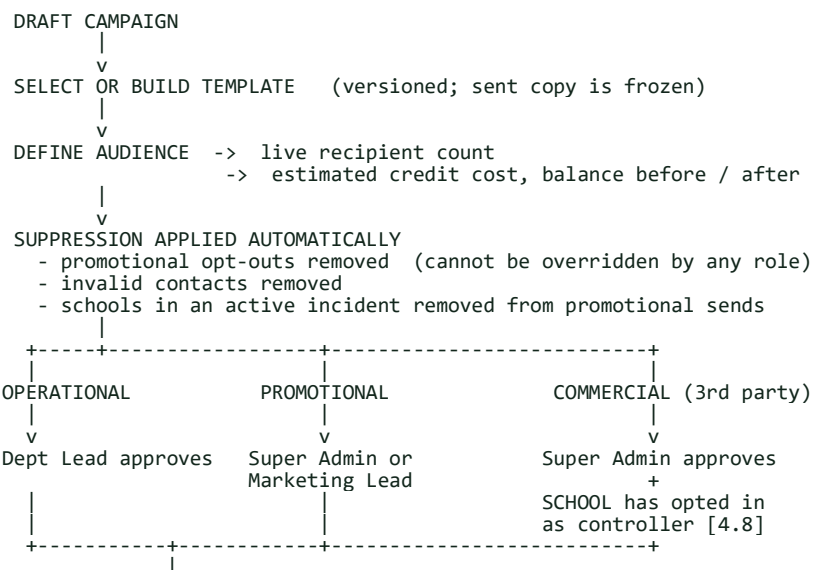
4.4 — Template Library

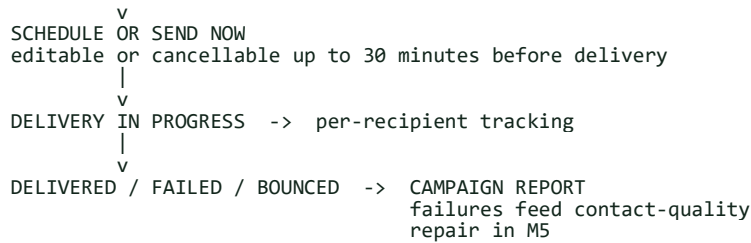
CATEGORY	CONTENTS
Onboarding	Welcome, provisioning confirmation, first-login guide, setup reminders
Trial pipeline	The full day 0 to day 37 sequence used by M3.10
Subscription	Renewal reminders, payment confirmation, trial expiry, suspension notice
Operational	Maintenance notice, feature announcement, term start reminder, incident notification
Compliance	Data processing agreement updates, privacy notice changes, breach notification
Migration	Migration started, records requiring attention, migration complete with reconciliation summary

- Approval status tracked per template, with the provider's rejection reason shown where a template is refused
- Only Super Admin may add or retire templates
- Every template is versioned. A message already sent always renders as the version that was sent, never as the current one.

4.5 — Campaign scheduling and send workflow

CAMPAIGN SEND — APPROVAL PATH DIFFERS BY COMMUNICATION TYPE





Campaign send workflow — suppression is applied before approval, so no approver ever sees a count that includes people who opted out.

4.6 — Delivery and Engagement Tracking

- Recipients targeted, delivered, and failed with reason: invalid number, opted out, carrier block, provider rejection
- Email open rate; WhatsApp read receipts where the provider supplies them; SMS delivery confirmations
- Failed contacts feed a contact-quality repair queue in M5, so that invalid numbers are corrected proactively rather than discovered during a results or renewal period

4.7 — Consent and Opt-Out

- Opt-in status stored per user, per channel, per communication type — not as a single global flag
- Opt-outs honoured immediately and permanently; the record of the opt-out is itself retained so that we can evidence having honoured it
- No role may override an opt-out on a promotional send. The override does not exist as a function, so it cannot be granted.
- Where consent was captured by the school rather than by us, the record stores which school captured it and when, because that is the chain we would have to evidence

4.8 — Third-Party Commercial Campaigns

THIS IS THE MOST EXPOSED PROPOSAL IN THE VERSION 3.0 DOCUMENT

Version 3.0 proposed that partner companies pay Future Realm to reach schools and parents, with a Sponsored label and a parent opt-in.

The problem is structural rather than presentational. When a school is on the platform, the school is the data controller for its students and guardians and Future Realm is the processor. A processor may only process controller data on the controller's documented instructions. Using a school's parent contact list to run our own advertising business is processing for our own purpose, not theirs — and a parent opt-in collected by us does not fix it, because we were never the party entitled to ask.

The label does not cure it either. Labelling a message Sponsored tells a parent who paid; it does not establish that we had the right to send it.

Recommendation. Remove third-party commercial campaigns to parents from version 1.0 entirely. If the revenue line is wanted later, it ships only in the form below, in which the school — not Future Realm — is the party deciding.

REQUIREMENT	SPECIFICATION
School opts in as controller	The school, through its own portal, explicitly enables third-party communications to its parent community, with a named category list. Silence is not consent, and it is not enabled by default.
Instruction is documented	The opt-in is recorded as a documented processing instruction under the data processing agreement, with version, acceptor and timestamp

REQUIREMENT	SPECIFICATION
Revenue is shared	The school receives a defined share. A school that is not paid is being used, and would be entirely right to see it that way.
Parents keep an independent opt-out	Parent opt-out overrides school opt-in, always, immediately
Withdrawal is one action	The school can withdraw at any time, from its own portal, taking effect on every scheduled campaign immediately
Category restrictions	No categories touching children's health, finance, religion or political content, and no offers directed at students rather than adults
Clear labelling	Sponsored labelling remains — necessary, but never sufficient on its own
Separate reporting	Commercial revenue tracked separately in M3 and separately disclosed to the school

THE COMMERCIAL ARGUMENT, NOT ONLY THE LEGAL ONE

The entire product is sold on the promise that the school owns its data and that we are the vendor who does not hold schools hostage. An advertising business running quietly on top of parent contact lists is the single fastest way to lose that positioning, and a competitor would only need to describe it accurately to damage us. The redesign above is not a concession; it is the only version that is consistent with what we are selling.

Key Data Entities

ENTITY	KEY FIELDS
Campaign	campaign_id, type, channel, audience_filter (json), template_id, template_version, scheduled_at, status, approved_by, school_instruction_ref (nullable)
Message Template	template_id, version, name, channel, category, language, body, placeholders[], approval_status, provider_template_ref, retired_at
Message Log	message_id, campaign_id, recipient_id, recipient_role, school_id, channel, delivery_status, failure_reason, opened_at, read_at
Consent Record	user_id, school_id, channel, communication_type, opted_in (bool), captured_by (school/platform), captured_at, opted_out_at
Audience Segment	segment_id, name, filter_criteria (json), recipient_count, last_calculated, created_by
Commercial Instruction	school_id, categories_permitted[], dpa_version, accepted_by, accepted_at, withdrawn_at (nullable), revenue_share_pct

Offline Behaviour

None internally. Scheduled sends are queued server-side and are unaffected by any internal user's connectivity — a campaign never depends on a browser tab staying open.

Notification Triggers

EVENT	RECIPIENT	CHANNEL
Campaign requires approval	Approving role	In-app plus email
Campaign send completed	Campaign creator	In-app, with the report attached
Delivery failure rate above threshold	Marketing Lead and CTO	In-app plus email
Template rejected by the provider	Super Admin	Email
School withdraws a commercial instruction	Marketing Lead and Account Manager	In-app, immediate

Permissions Note

- Send platform-wide operational messages: Department Lead approval required
- Send promotional campaigns: Super Admin or Marketing Lead
- Send third-party commercial campaigns: Super Admin, and only to schools holding a current documented instruction
- Create or manage templates: Super Admin only
- View delivery reports: Marketing Lead and above, scope-narrowed
- Override an opt-out: not permitted for any role. The function does not exist.

User Stories

- As a product manager, I want to email all school admins about a new release, so that adoption starts immediately across every school without individual outreach.
- As an operations lead, I want to schedule a renewal reminder that fires ten days before each school's term ends, so that no renewal is missed through human oversight.
- As a marketing lead, I want the recipient count and credit cost before I confirm a send, so that I never discover the cost of a campaign after sending it.
- As a compliance-conscious team member, I want opted-out recipients excluded automatically from every future campaign, with no override available to anyone, so that Future Realm never damages trust by contacting someone who said no.
- As a school proprietor, I want to decide myself whether any third party may ever reach my parents, so that my parent community is mine to protect, not my vendor's audience to sell.

MODULE 5

Cross-Platform User Management

PURPOSE

Visibility and control over every individual account across every school. It exists to resolve account problems schools cannot resolve themselves, to maintain security across all accounts, and to give Future Realm complete accountability over who is using the platform.

It is also where the platform's identity model — one account, many school memberships — is administered.

CHANGED IN V4.0 *Multi-school membership made explicit throughout. Individual Users Register added as 5.9. Support access rewritten in 5.10 so the school can see when we enter their portal. Contact-quality repair added as 5.11.*

5.1 — Global User Registry

- Searchable by name, email, phone, user identifier, role, school and account status
- Columns: Name, Identifier, Role, School Memberships, Last Login, Status, Last Device
- A user with memberships at several schools appears once, not once per school. Anything else creates the duplicate-account problem the identity model exists to prevent.
- Scope narrows the registry: an Assigned-scope Support Agent sees only users belonging to schools on their open tickets

5.2 — User Profile

- Personal details, account status, creation date, and platform identifier alongside each school identifier held
- Last ten login sessions with timestamp, device, browser and approximate location
- Role-specific activity: teachers — score entries and attendance marked this term; guardians — notifications received and opened, last result viewed; school admins — administrative actions and modules accessed
- Linked records: the students a guardian is connected to, across every school those children attend, with each school's data shown separately and never merged into one view
- Support access history — every session in which a Future Realm employee viewed the platform as this user

5.3 — Account Actions

ACTION	WHEN USED	MINIMUM ROLE
Force password reset	User locked out and the school admin cannot resolve it	Support Agent
Suspend account	Suspected unauthorised access or policy violation	Support Lead
Reinstate account	Reverses a suspension after resolution	Support Lead
End all active sessions	Compromise suspected; takes effect immediately everywhere	Support Lead
Remove one school membership	A person has left one school but remains at another	Support Lead, with school admin confirmation
Deactivate account permanently	Only on written request from the school	Super Admin

All actions are written to M10 immediately, and the affected user is notified of anything that changes their access.

5.4 — Multi-School Membership Administration

- A single account may hold memberships at several schools: a teacher working at two, a guardian with children at different schools, an external examiner, a proprietor owning several schools
- Each membership carries its own role, permissions and record access, independently
- Leaving one school does not affect any other membership
- Audit entries always record which school membership an action occurred under
- **No query, search, report or export may return data spanning two schools.** This is enforced by the data-scope layer described in Section A, not by careful interface design — including for internal staff.

5.5 — Suspicious Activity Detection

SIGNAL	THRESHOLD
Failed login attempts	More than 10 in one hour
Login from an unrecognised country or city	Any
Simultaneous sessions from two distant locations	Any
Mass data export from a single session	Above a configured record count
Permission changes outside normal hours	Any, for school admin accounts
Bulk guardian contact changes	Above a configured count in one session — a known pattern in account takeover

Flagged accounts enter a review queue with three actions: dismiss the flag, force a password reset, or suspend pending investigation. Every outcome is recorded, including dismissals — a dismissed flag that later turns out to matter must be traceable to a decision and a person.

5.6 — Duplicate Account Detection

- Automatic detection on shared phone number, shared email, or the same name linked to the same school
- Flagged pairs enter a duplicate review queue
- Admin can merge, dismiss, or escalate to the school admin
- Merging requires the user's consent where both accounts have been used. Taking over a person's account without asking is the one place the identity model refuses to prioritise speed.

5.7 — Contact Quality Repair

New in v4.0. Delivery failures from M4.6 collect here rather than evaporating into a campaign report.

- Queue of users whose email or phone has failed delivery repeatedly, grouped by school
- Routed to the school for correction, since the school holds the relationship and the correct data — we never edit a guardian's contact details ourselves
- A school with a high proportion of unreachable guardians is surfaced to its Account Manager, because a school whose parents cannot be reached is not receiving the product's core value

5.8 — School Admin Account Recovery

The most common support scenario: a school admin locked out with no access to their email.

VERIFIED IDENTITY RECOVERY

ADMIN REPORTS LOCKOUT

↓

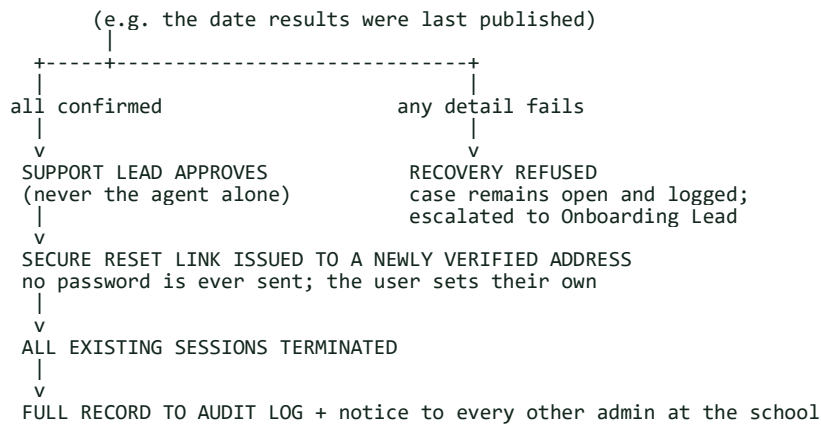
SUPPORT AGENT OPENS A RECOVERY CASE (reason recorded before anything else)

↓

PHONE VERIFICATION AGAINST THE REGISTERED NUMBER

caller must confirm ALL of:

- school name and registered address
- proprietor name
- the registered primary phone number
- at least one recent system activity detail



Account recovery — the last line is the control that matters: the school's other administrators are always told.

5.9 — Individual Users Register

NEW IN V4.0

Teachers who sign up on their own hold accounts that belong to no school. The strategy identifies these as a ranked lead list — a route into schools that cold outreach cannot open. Version 3.0 had nowhere to put them.

- Every self-registered individual account, with the school they named as an unverified claim, their subjects and level, registration date and activity level
- Sorted by cluster strength: how many individual accounts have named the same school. Three teachers naming one school is a materially warmer lead than three teachers at three schools.
- Cluster view shows named school, count of individuals, their combined activity, whether the school already exists on the platform, and whether a partner has registered a deal against it
- Routing to a partner happens only under a registered deal in M13, so that attribution is established before the introduction rather than argued about afterwards

THE CONSTRAINT THAT GOVERNS THIS ENTIRE REGISTER

The free teacher tier is being designed to be structurally incapable of holding identifiable student data — no full names, photographs, dates of birth, admission numbers or guardian contacts. This register must therefore never surface student-level data of any kind from an individual account, in any view, export or aggregate. What it shows is the teacher and the school they claim to work at. Nothing about the children.

Stated here because a lead-generation surface is exactly where the temptation to enrich a record with more detail arrives, and it must be closed in the specification rather than in a code review.

5.10 — Support Access (View-As) and the School's Record of It

A support agent may view the platform exactly as a specific user sees it, in order to diagnose a reported problem rather than guess from a description.

CONTROL	SPECIFICATION
Default mode	Read-only. No action may be taken on the user's behalf.
Elevation to act	Super Admin escalation, plus explicit confirmation from a school administrator, recorded before the session

CONTROL	SPECIFICATION
Maximum duration	30 minutes, expiring automatically regardless of activity
Reason	Recorded before the session starts. A session cannot begin without one.
Session logging	Start, end, every page viewed and every action taken, written to M10
Blocked contexts	No support access into a school with an unresolved dispute, and none into a guardian account with a safeguarding restriction on the linked student

THE CORRECTION: THE SCHOOL MUST BE ABLE TO SEE IT TOO

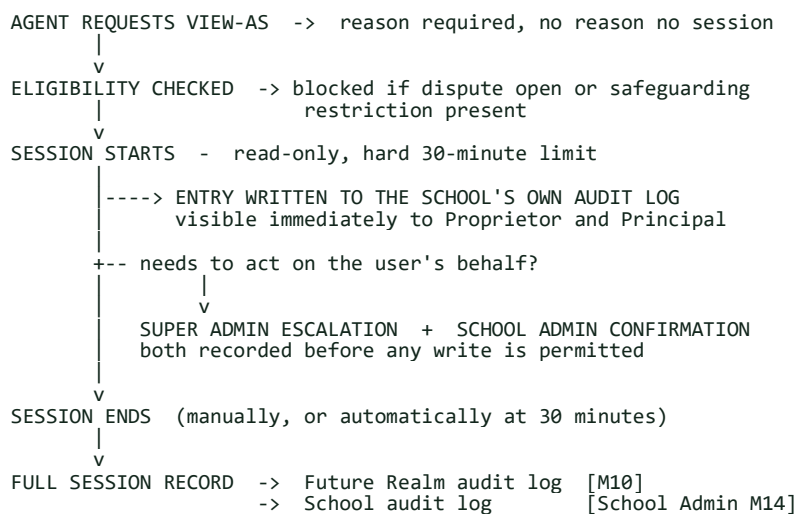
Version 3.0 logged every session to our own audit trail and told the school nothing. That is below the standard of the platforms we are benchmarking against — enterprise software has moved decisively towards making vendor access visible to the customer, and often disableable by them.

The exposure is sharper for us than for a generic business tool, because the portal we are entering holds children's records and the school, not Future Realm, is the controller of them.

From v4.0: every support access session writes a matching entry into the school's own audit log (School Admin BRD, Module 14), visible to the Proprietor and Principal, showing who accessed, when, for how long, why, and whether it was read-only. Anything beyond read-only requires a school administrator's recorded confirmation first.

This is also a sales asset. "You can see every time we look at your data, and we cannot change anything without your say-so" is a sentence no competitor in this market can currently say.

SUPPORT ACCESS SESSION



Support access — dual-logged by design, to us and to the school.

Key Data Entities

ENTITY	KEY FIELDS
Platform User	user_id, platform_reference, name, email, phone, status, last_login, device_last_used, created_at
School Membership	membership_id, user_id, school_id, school_identifier, role, permission_set_id, data_scope, joined_at, left_at (nullable)

ENTITY	KEY FIELDS
Individual Account	user_id, claimed_school_name, claimed_location, subjects[], level, verified_claim (bool), cluster_key, last_active
Support Access Session	session_id, admin_id, target_user_id, school_id, reason, mode (read_only/elevated), escalated_by, school_confirmed_by, started_at, ended_at, pages_viewed[], actions_taken[]
Suspicious Activity Flag	flag_id, user_id, flag_type, detected_at, reviewed_by, admin_action, resolved_at
Duplicate Flag	flag_id, user_id_a, user_id_b, match_criteria, status, consent_obtained (bool)
Account Recovery Record	recovery_id, user_id, agent_id, verification_details_confirmed[], approved_by, new_email, completed_at, outcome
Contact Quality Issue	issue_id, user_id, school_id, channel, failure_count, first_failed_at, routed_to_school_at, resolved_at

Offline Behaviour

None internally. A support access session requires connectivity by design — a cached view of a user's portal would be a stale representation of children's data sitting on a staff laptop.

Notification Triggers

EVENT	RECIPIENT	CHANNEL
Support access session started	School Proprietor and Principal	In-app entry in their audit log; email digest daily
Elevated (write-capable) support access requested	School administrator	In-app plus email, requiring confirmation
Account suspended or reinstated	The user, and their school admin	Email
Password reset forced	The user, and every other admin at the school	Email
Suspicious activity flagged	Support Lead	In-app
Recovery completed	Every other admin at the school	Email

Permissions Note

- View global user registry: Support Agent and above, narrowed by scope
- Force password reset: Support Agent and above
- Suspend or reinstate: Support Lead and above
- Initiate support access: Support Agent and above — always logged, no exceptions, no suppression
- Elevate support access to write: Super Admin, plus school administrator confirmation
- Account recovery to a new email: Support Lead approval, never an agent alone
- Deactivate permanently: Super Admin only
- Individual Users Register: Sales and Onboarding roles, and above. Student-level data is not available to any role because it does not exist.

User Stories

- As a support agent, I want to search for any user by phone number across the platform, so that I can resolve their issue without first asking them which school they belong to.
- As a support agent, I want to briefly view a teacher's portal as they see it, so that I can diagnose a reported bug rather than guessing from a text description.
- As a school proprietor, I want to see in my own audit log every time Future Realm staff viewed my school, so that I can trust the vendor holding my students' records because I can verify what they did.

- As a security-conscious admin, I want to suspend a compromised guardian account immediately, so that a security issue never waits for a full school-level response.
- As a data quality lead, I want duplicate guardian accounts flagged automatically, so that a family is never confused by two logins linked to the same child.
- As a sales lead, I want self-registered teachers ranked by how many named the same school, so that I approach the school where three teachers are already asking for us rather than one where nobody is.

MODULE 6

Support & Ticketing

PURPOSE

Ensures every school's problem is seen, owned, tracked and resolved. Support quality is one of the strongest drivers of retention in this market: a school that gets fast, competent, human support stays and tells other proprietors; one that feels ignored leaves and tells them too.

This module makes support excellence systematic and measurable rather than dependent on who happens to be paying attention.

CHANGED IN V4.0 *Migration and onboarding categories added. Data correction workflow tightened to require school authorisation as well as internal approval. Service level definitions made time-zone aware.*

6.1 — Unified Ticket Inbox

- WhatsApp conversations logged by the agent; emails parsed into tickets automatically; in-app support form creating tickets directly
- No support conversation may exist only in a personal WhatsApp thread. Every interaction is logged here, because a conversation that lives on one employee's phone is lost the day they leave.

6.2 — Ticket Registry

- Columns: Ticket, School, Contact, Category, Priority, Assigned Agent, Status, Opened, Service Level Deadline, Service Level State
- Default view: open tickets by priority, then by deadline
- Board view with columns New, Triaged, In Progress, Awaiting School, Resolved, Closed
- Filter by priority, category, agent, service level state, school, and tier

6.3 — Priority and Service Levels

PRIORITY	TARGET	DEFINITION	ESCALATION
Critical	1 hour	Result or data visibility issue, login failure, data loss	Immediate alert to Support Lead
High	4 hours	Core workflow broken — results, fees, attendance — with no workaround	Alert at 2 hours if unresolved
Standard	8 hours	Feature not behaving as expected; a workaround exists	Alert at 6 hours
Low	24 hours	How-to question, minor inconvenience, feature request	Alert at 20 hours

- Targets are measured in the school's own working hours in the school's own time zone, not ours. A four-hour target that expires overnight in Nairobi while our office is closed is a promise we cannot keep and should not make.
- Working hours per country are a configuration record, alongside public holidays

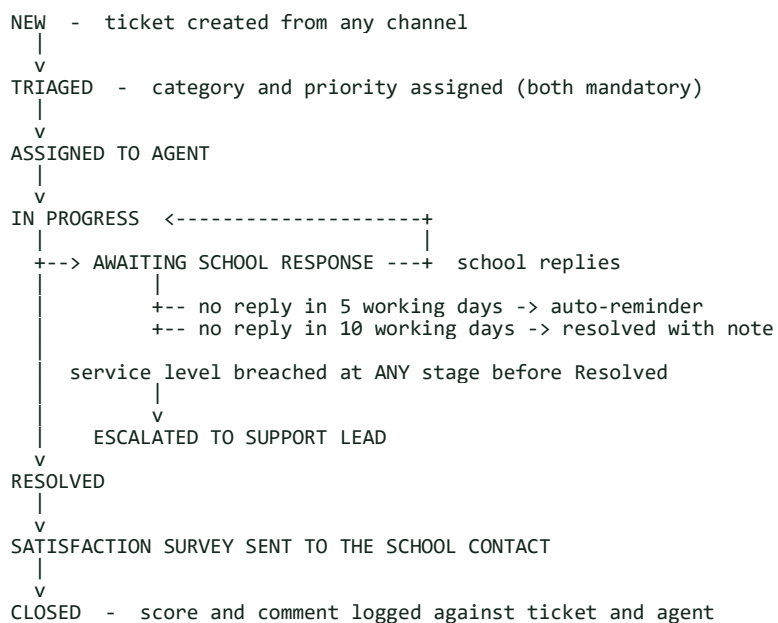
6.4 — Issue Categories

CATEGORY	COVERS
Login and Access	Cannot log in, forgotten password, wrong permissions

CATEGORY	COVERS
Result Computation	Incorrect grade, missing score, result not publishing
Fee Management	Incorrect balance, payment not reflecting, receipt error
Notification Delivery	Message not sending or not received on any channel
Sync and Offline	Data not syncing, offline records missing after reconnection
Billing and Subscription	Invoice query, payment not confirmed, tier change request
Data Correction	Name error, wrong student in class, published result error
Data Migration	Import problem, records rejected, reconciliation dispute — new in v4.0
Onboarding and Setup	Setup wizard, configuration, template selection — new in v4.0
Feature Question	How-to, training request, configuration guidance
Bug Report	Behaviour not matching expected behaviour
Other	Must be reclassified before the ticket can be closed

6.5 — Ticket lifecycle

TICKET LIFECYCLE — CREATION TO SATISFACTION-VERIFIED CLOSURE

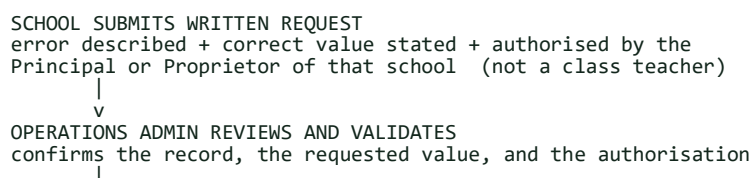


Ticket lifecycle — with an escalation path from every state before resolution.

6.6 — Data Correction Requests

A specific sub-process for any change to historical student data — the category with the greatest potential to cause harm and the greatest need for evidence afterwards.

DATA CORRECTION — NO UNLOGGED CHANGE TO A CHILD'S RECORD, EVER



```

      v
IMPACT ASSESSED
does this change a published result, a computed position,
a fee balance, or anything already sent to a parent?
      |
      +-- yes --> downstream artefacts identified and listed
                  (report cards, broadsheets, notifications)
      |
      v
SUPER ADMIN APPROVES
      |
      v
CORRECTION APPLIED - before value, after value, approver,
requester, authorising school officer, reason, timestamp
all written to the audit log as one immutable entry
      |
      v
DOWNSTREAM ARTEFACTS REISSUED where required
      |
      v
SCHOOL NOTIFIED WITH A CORRECTION SUMMARY
and the record appears in the school's own audit log

This process cannot be bypassed by any role at any level.

```

Data correction — school authorisation, internal approval, downstream impact, and a permanent record.

WHY SCHOOL AUTHORISATION WAS ADDED IN V4.0

Version 3.0 required internal validation and Super Admin approval, but nothing from the school beyond a request. Since the school is the controller of the record being changed and we are the processor, a correction made on a request from someone with no authority at the school is us altering a child's record on our own initiative. Requiring the Principal or Proprietor's authorisation costs one email and removes the entire exposure.

6.7 — Canned Responses and Knowledge Base

- Pre-written responses per category, personalised before sending
- Internal knowledge base documenting how to diagnose recurring issues
- Maintained by the Support Lead from observed ticket patterns, with the top recurring categories reviewed monthly
- Where a knowledge article is used more than a set number of times in a month, it is raised as a product defect candidate to M7.8 — recurring support volume is usually a design problem wearing a support costume

6.8 — Satisfaction and Quality

- One-question satisfaction survey after every resolved ticket, on a five-point scale with an optional comment
- Score logged against both ticket and agent
- Per-agent averages visible to the Support Lead in M7
- Consistently low scores trigger a coaching review, not an automated consequence — the sample sizes here are far too small for anything else to be fair

6.9 — Support Analytics

- Tickets opened and resolved within service level
- Average resolution time by priority
- Category breakdown, ranked
- Per-agent volume and satisfaction
- Repeat issue rate — the same school raising the same category more than once in a term
- Ticket volume per hundred schools, by tier — the number that tells us whether support cost scales with the business or faster than it

Key Data Entities

ENTITY	KEY FIELDS
Ticket	ticket_id, school_id, user_id, channel, category, priority, status, assigned_to, sla_deadline, sla_timezone, created_at, resolved_at, closed_at
Ticket Note	note_id, ticket_id, author_id, note_text, is_internal (bool), tagged_users[]
Satisfaction Response	ticket_id, score, comment, submitted_at
Canned Response	response_id, category, title, body, language, use_count, last_updated
Data Correction Record	request_id, ticket_id, school_id, authorised_by_school_officer, entity_type, entity_id, field, old_value, new_value, downstream_artefacts[], approved_by, completed_at

Offline Behaviour

None internally. On the school side, the in-app support form must queue offline and submit on reconnection — a school reporting that its data will not sync is by definition a school with a connectivity problem, and a support form that needs connectivity to report it is a design failure.

Notification Triggers

EVENT	RECIPIENT	CHANNEL
Ticket created	School contact and assigned agent	Email plus in-app
Critical ticket created	Support Lead	In-app plus email, immediately
Service level breach	Support Lead	In-app plus email
Awaiting school response for 5 days	School contact	Email plus WhatsApp
Ticket resolved	School contact	Email, with the satisfaction survey
Data correction completed	School admin and the authorising officer	Email with correction summary

Permissions Note

- View and update tickets: Support Agent and above, scoped to assigned tickets unless elevated
- Reassign across agents: Support Lead and above
- Override a service level state: Support Lead and above, with a reason
- Approve data corrections: Super Admin only
- View per-agent performance: Support Lead and above

User Stories

- As a support agent, I want every incoming ticket timestamped against our service level in the school's own time zone, so that I always know which tickets are at genuine risk rather than which are merely old in my time zone.
- As a support lead, I want critical tickets visually distinct from routine questions, so that my team's attention goes to the highest-stakes issues first.
- As a customer success manager, I want a school's full support history before a renewal call, so that I can address recurring frustrations rather than being surprised by them.
- As a Super Admin, I want every data correction to require the school's own authorisation as well as mine, so that we can demonstrate that we never altered a child's record on our own initiative.
- As a school principal, I want to be told exactly what was changed and what was reissued after a correction, so that I can explain the change to a parent without contacting support again.

MODULE 7

Platform Analytics & Business Intelligence

PURPOSE

The single source of truth for every metric that measures whether the product and the business are working. No strategic decision — what to build next, which market to enter, which schools to prioritise — should be made without examining this module first.

It exists to make honest measurement easier than optimistic measurement.

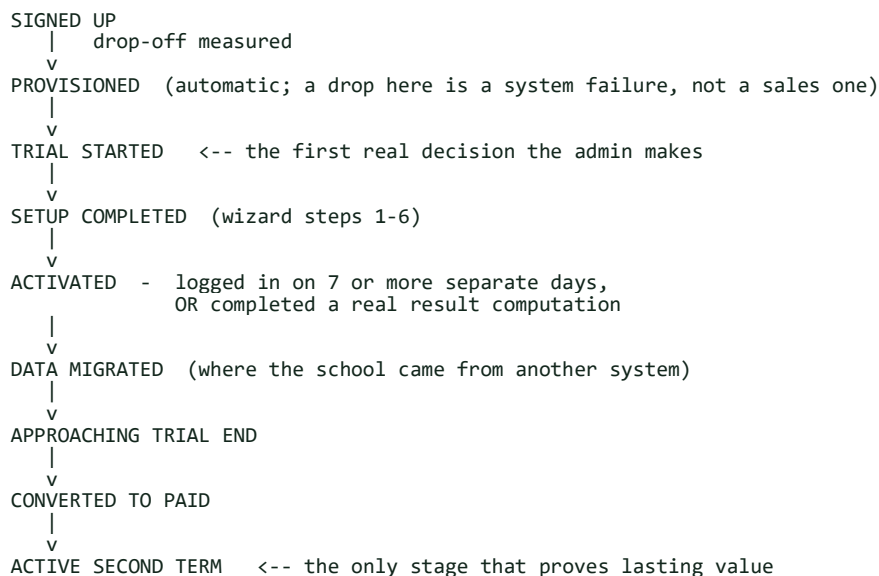
CHANGED IN V4.0 *Displacement and migration metrics added, because the market strategy is a displacement play and nothing was measuring it. Cohort analysis extended to acquisition source. Small-cohort suppression added for scope-limited users.*

7.1 — Product Adoption Heatmap

- Module-by-module grid of average weekly usage across all active schools: high (used three or more times a week), medium (one to two), low (not used at all), not applicable (not on this tier)
- Segmentable by tier, size band, region and tenure — a module ignored by small schools and loved by large ones is a very different finding from one nobody uses
- Directly identifies which features drive engagement and which are dead weight, and is the primary input to the next development sprint

7.2 — Growth and Funnel Reporting

CONVERSION FUNNEL, WITH DROP-OFF MEASURED AT EVERY STAGE



Funnel — every stage clickable to the list of schools currently sitting in it.

- Week-on-week movement tracked so the team can see whether conversion is improving, not merely what it is
- Funnel segmented by acquisition source, so partner-sourced schools can be compared with self-signup and direct sales

7.3 — Cohort Retention

- Schools grouped by the term they joined, showing the percentage still active one, two and three terms later
- The most honest measure available of whether the product creates lasting value. Eighty per cent at three terms means it is working; forty per cent means there is a retention problem that must be solved before any further scaling.
- Cohorts also cut by acquisition source, tier and city — a partner channel with strong acquisition and weak retention is a cost, not a channel, and the aggregate figure will hide it

7.4 — Churn Analysis

When a school's status changes to Closed, the handling admin must log a reason from a defined list. The ticket cannot be closed without one.

REASON	WHAT IT TRIGGERS
Price too high	Pricing strategy review
Switched to competitor	Competitive intelligence — which competitor, and what won it
School closed	External; no action
Product issues	Product roadmap input, linked to the specific tickets
Insufficient support	Support quality review, linked to that school's ticket history
Low staff adoption	Onboarding and training review
Migration failed or incomplete	New in v4.0 — routes to M14 as a process defect
Other	Must be described. Cannot be closed without detail.

Monthly churn reason breakdown is reviewed in the team strategy meeting. Where any single category exceeds a quarter of total churn, it escalates to a named response plan with an owner.

7.5 — Revenue and Financial Reporting

- Recurring revenue trend, gross and net of commission
- Revenue by tier and, separately, by school size band
- Average revenue per school over time
- Lifetime value by tier, calculated from observed durations
- Notification credit revenue as a proportion of total
- Cost to serve per school — infrastructure, messaging, support hours — so that margin by tier is a measured figure rather than an assumed one

7.6 — Geographic and Segment Performance

- Distribution and revenue by country, state and city, shown as ranked table and map
- Density indicator per city: the point at which proprietor word-of-mouth begins to compound, which is the trigger for opening the next city rather than the next state
- Performance by size band and curriculum type

7.7 — Displacement and Competitive Intelligence

NEW IN V4.0

The market strategy states plainly that this is a displacement play rather than a discovery play: roughly half of the genuinely addressable segment already owns something. Nothing in v3.0 measured displacement at all, which meant the central strategic assumption of the business was unfalsifiable.

- Previous system recorded at signup and at migration, as a selectable list plus free text
- Displacement rate: proportion of new schools arriving from another system rather than from paper

- Win analysis by displaced system — which competitor we take schools from, and what they said made them move
- Migration completion rate by source system, which tells us where our import tooling is weak and therefore which competitor we are effectively unable to displace
- Loss analysis: schools that trialled and returned to their previous system, with the reason

7.8 — Feature Request Intelligence

- Feature requests tagged from support tickets and matched by keyword to count how many schools asked for the same thing
- Priority score from request frequency, tier of requesting schools, and associated ticket severity
- Top ranked requests shown as a table, as the first reference for roadmap planning
- Requests arising from churned schools weighted separately and visibly — the feature that would have retained a school we already lost is worth knowing about, and an aggregate ranking buries it

7.9 — Custom Report Builder

- Reports built from any combination of available metrics and dimensions
- Filter by any dimension: tier, region, size, tenure, curriculum, acquisition source, partner
- Export to Excel and CSV in one action
- Saved reports reusable for recurring reporting — investor updates, board packs, partner statements
- Small-cohort suppression applies for scope-limited users: a segment containing fewer than a configured number of schools returns the aggregate only, without drill-through, so that a chart cannot be used to identify an individual school

Reporting cadence

METRIC FAMILY	CADENCE
Adoption heatmap	Daily rollup, weekly summary
Funnel and conversion	Real-time capture, weekly summary
Cohort retention	Weekly recalculation
Revenue	Daily rollup, monthly summary
Churn analysis	Weekly recalculation
Satisfaction and promoter score	Per survey cycle, termly
Geographic performance	Weekly rollup
Displacement	Monthly

Key Data Entities

ENTITY	KEY FIELDS
Metric Snapshot	metric_name, dimension, dimension_value, value, period, calculated_at
Funnel Event	school_id, funnel_stage, entered_at, acquisition_source, partner_id
Cohort Record	cohort_term, school_id, acquisition_source, active_at_term_1, active_at_term_2, active_at_term_3
Churn Record	school_id, churned_at, reason, competitor_named, logged_by, notes
Displacement Record	school_id, previous_system, migrated (bool), migration_completed_at, outcome
Satisfaction Response	school_id, survey_type, score, comment, submitted_at
Feature Request Tag	request_id, source_ticket_id, keyword, school_id, tier, school_churned (bool), submitted_at
Custom Report	report_id, created_by, filters (json), export_format, generated_at

Offline Behaviour

None. Analytics are computed server-side from authoritative data and are never derived on a client.

Notification Triggers

EVENT	RECIPIENT	CHANNEL
Weekly business summary	Founders and Department Leads	Email, Monday morning
Churn category exceeds threshold	Super Admin and the relevant Department Lead	Email
Cohort retention falls below target	Founders	Email
City reaches density threshold	Growth Lead	In-app plus email

Permissions Note

- Full analytics: Department Lead and above
- Revenue and financial reports: Finance Lead and Super Admin
- Support metrics: Support Lead and above
- Funnel and growth: Sales, Onboarding and Marketing Leads and above
- Satisfaction and promoter score: Customer Success Lead and above
- Custom report builder: Department Lead and above
- Scope None users receive aggregates only, with drill-through disabled and small-cohort suppression applied

User Stories

- As a product manager, I want to see which modules are used weekly and which are ignored, so that I know where to invest the next development sprint rather than guessing.
- As a founder, I want the trial-to-paid funnel with drop-off identified at every stage, so that I know exactly which stage of onboarding is losing us schools.
- As a growth lead, I want performance broken down by city with a density indicator, so that I know when Lagos is dense enough to justify opening the next city.
- As a founder, I want cohort retention cut by acquisition source, so that I can tell an acquisition channel from a leaky bucket.
- As a strategy lead, I want displacement measured rather than assumed, so that our central market thesis is something we can test instead of something we repeat.
- As a founder, I want to export a clean revenue and growth report in two minutes, so that fundraising conversations never wait on engineering.

MODULE 8

Feature Configuration & Tier Management

PURPOSE

Central control over what each subscription tier can access, with safe staged feature rollout, per-school overrides, and management of branding assets — none of it requiring a code deployment.

The governing idea is that commercial packaging and product capability are separate things, and the boundary between them belongs in data.

CHANGED IN V4.0 Rollout process extended with explicit exit criteria and stale-flag detection. Overrides given mandatory expiry and audit. Tier matrix marked as commercially provisional alongside pricing.

8.1 — Tier and Feature Matrix

A grid of every capability against every tier. Super Admin may update it at any time; changes apply to new subscriptions and upgrades, and never retroactively alter an existing school's access unless explicitly actioned.

CAPABILITY	STARTER	STANDARD	ELITE
Student registration and records	Yes	Yes	Yes
Attendance management	Yes	Yes	Yes
Result computation and report cards	Yes	Yes	Yes
Fee management (manual recording)	Yes	Yes	Yes
Email notifications	Yes	Yes	Yes
Parent, teacher and student portals	Yes	Yes	Yes
Basic analytics	Yes	Yes	Yes
Full data export	Yes	Yes	Yes
SMS notification credits	—	Yes	Yes
WhatsApp notification credits	—	Yes	Yes
Payment integration (Phase 2)	—	Yes	Yes
Computer-based examination module	—	Yes	Yes
Staff payroll	—	—	Yes
Multi-branch support (Phase 2)	—	—	Yes
Predictive and at-risk student analytics	—	—	Yes
Biometric attendance integration	—	—	Yes
Custom report card branding	—	—	Yes
Regulatory reporting exports	—	—	Yes
Programmatic access for integration	—	—	Yes
Dedicated account manager	—	—	Yes

FULL DATA EXPORT IS DELIBERATELY ON EVERY TIER, INCLUDING STARTER

It would be commercially tempting to make export an upgrade trigger. It is the one capability that must never be tiered, because the promise that a school owns its data absolutely is the promise the whole displacement strategy rests on. A school that can only extract its records by paying more is exactly the situation we are selling against.

8.2 — Per-School Feature Override

- Sales may request an override; they cannot apply one
- Requests route to the Product Lead for approval
- Every override carries a mandatory expiry date. It may be renewed but never left open-ended, because an override with no expiry is a pricing decision made by accident.
- Common uses: demonstrating upgrade value during a trial; correcting a billing error where a school paid for a higher tier but remained on a lower one
- Active overrides are listed with days remaining, and a weekly digest goes to the Product Lead

8.3 — Feature Flag Rollout**STAGED ROLLOUT WITH AN INSTANT ROLLBACK PATH AT EVERY STAGE**

```

FEATURE BUILT -> FLAG CREATED, OFF BY DEFAULT
    |
    v
INTERNAL ONLY - Future Realm demo schools
    |               exit criteria: no critical defects for 3 days
    |
    v
PILOT GROUP   - 5 to 10 named schools who know they are piloting
    |               exit criteria: 7 days minimum, no critical defects,
    |               support volume not materially raised, feedback reviewed
    |
    v
PARTIAL ROLLOUT - 25 per cent of schools
    |               exit criteria: 7 days minimum, error and sync rates
    |               within normal range for the cohort
    |
    v
FULL PLATFORM ROLLOUT
    |
    v
FLAG RETIRED - the feature becomes ordinary product

AT ANY STAGE: issue detected
    |
    v
INSTANT ROLLBACK - flag disabled platform-wide in seconds,
                  CTO or Product Lead, no deployment required, incident opened [M10.3]

```

Feature rollout — no new capability reaches every school on the same day.

- Exit criteria are recorded against the flag, not held informally. A stage cannot be advanced without them being marked met, and by whom.
- Any flag in pilot or partial state for more than 30 days is flagged for review. A permanent flag is technical debt that eventually behaves unpredictably.
- Retiring a flag is a tracked task, not an optional tidy-up
- Schools in a pilot group are told they are piloting. Quietly experimenting on a paying school's live data is not a defensible practice when that data is children's records.

8.4 — Branding Configuration (Elite)

- School uploads a logo and brand colours through its own portal
- Admin reviews for quality and appropriateness, and previews against sample data before applying
- Applied assets are versioned, so a reissued historical report card renders with the branding that was current at the time it was originally issued

Key Data Entities

ENTITY	KEY FIELDS
Feature	feature_id, name, module, description, default_tier_access, retired_at
Feature Override	override_id, school_id, feature_id, override_status, expiry_date, requested_by, approved_by, reason
Feature Flag	feature_id, rollout_stage, rollout_pct, pilot_school_ids[], exit_criteria (json), criteria_met_by, stage_entered_at, rollback_count
Branding Asset	asset_id, school_id, version, logo_ref, primary_colour, secondary_colour, applied_to, approved_by, applied_at, superseded_at

Offline Behaviour

None internally. On the school side, the currently effective flag state must be cached with the application so that an offline device does not lose access to a feature it had a minute earlier, and a rollback must apply on the next successful sync.

Notification Triggers

EVENT	RECIPIENT	CHANNEL
Override approaching expiry (7 days)	Requesting sales user and Product Lead	In-app plus email
Flag rolled back	CTO, Product Lead, Support Lead	In-app plus email, immediate
Flag stale beyond 30 days	Product Lead	Weekly digest
School added to a pilot group	School admin	Email — they are told, always
Branding assets applied	School admin	Email with preview

Permissions Note

- View the tier and feature matrix: all internal roles
- Edit the matrix: CTO, Product Lead and Super Admin only
- Request an override: Sales and above
- Approve and apply an override: Product Lead and Super Admin
- Create, advance or roll back a flag: CTO, Product Lead and Super Admin
- Manage branding assets: Operations Admin and above, with Super Admin confirmation before applying

User Stories

- As a product manager, I want to enable a feature for ten pilot schools before releasing it platform-wide, so that a critical defect is found on ten schools rather than on all of them.
- As a sales lead, I want to request temporary access to a higher tier's features during a demonstration, so that I can show upgrade value without a developer.
- As a CTO, I want to disable a feature platform-wide instantly, so that an issue is contained in minutes rather than hours.
- As a school proprietor, I want to be told when my school is in a pilot group, so that nobody experiments on my students' records without telling me.
- As a product lead, I want every override to carry an expiry, so that we never discover a school has been on the wrong tier free of charge for two years.

MODULE 9

Academic & Curriculum Configuration Library

PURPOSE

The shared building blocks every school draws from when configuring its own instance of the platform: curriculum structures, subject frameworks, assessment models, grading scales, calendars and report card layouts.

This module is where the global expansion strategy either succeeds or fails. If entering Ghana, Kenya or the Philippines requires a configuration record, this module was designed correctly. If it requires an engineering sprint, it was not — and by then the cost of fixing it is a rewrite of the result computation engine.

CHANGED IN V4.0 *Rewritten and greatly expanded. Version 3.0 described a template library in nine lines and did not specify what a template actually contains. It also described the previous Nigerian curriculum structure, which was superseded by the national curriculum that took effect from the 2025/2026 session. This version specifies the full configuration model, the Nigerian reference implementation in detail, template versioning for schools caught mid-transition, and the portability tests every future template must pass.*

9.1 — The governing principle, stated precisely**EVERYTHING CULTURAL IS CONFIGURATION, NEVER CODE**

A curriculum is not a feature. It is a data structure describing how one education system organises levels, subjects, assessment and grading. The platform's job is to hold that structure faithfully and compute correctly against it — not to know anything about Nigeria in particular.

The practical test is blunt. If a developer would have to open an editor to launch a new country, this module has failed. Adding Ghana must mean inserting configuration records and nothing else.

Three consequences follow, and each is a hard constraint on every other module:

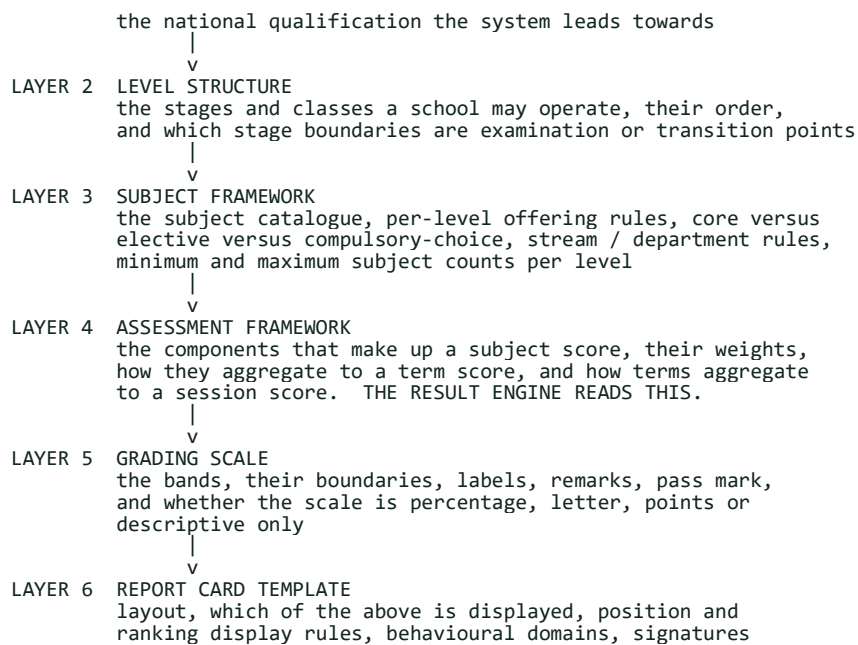
- No module may contain a literal reference to a Nigerian grade band, subject name, term count or class level. Where one is needed, it is read from the school's active curriculum configuration.
- The result computation engine takes the assessment framework as an input, never as an assumption. It must produce correct results for a two-semester grade-point system with the same code path it uses for a three-term percentage system.
- Master templates are owned by Future Realm; the copy a school customises is owned by that school. A change to a master template never reaches back into a school's live configuration, because a silent change to a grading band would silently change every result computed after it.

9.2 — The six-layer curriculum configuration model

Every curriculum on the platform, anywhere in the world, is expressed as six layers. Each layer is independently replaceable, which is what makes hybrid cases possible — a Nigerian school running a British curriculum on a Nigerian calendar, for example, which is common among international schools in Lagos and Abuja.

THE SIX LAYERS, AND WHAT EACH ONE GOVERNS

LAYER 1 EDUCATION SYSTEM (country level)
calendar shape, term or semester count, week start, currency,
language, date format, phone format, promotion policy,



A school selects a Layer 1 template at signup. Layers 2-6 are pre-populated from it, and every one of them can then be replaced independently by that school without affecting the others.

The six layers — independently replaceable, which is what makes hybrid and international schools work.

9.2.1 — Layer 1: Education System

FIELD	PURPOSE	NIGERIAN VALUE	CONTRAST CASE
Calendar shape	Whether the year is divided into terms or semesters	Three terms	United States: two semesters. Philippines: two semesters with a summer term.
Session naming	How an academic year is written	2026/2027	United States: 2026–27. Australia: single calendar year, since the year starts in January.
Session start month	When the academic year begins	September	Australia and South Africa: January. India: April or June by state.
Week start	First day of the school week	Monday	Retained as configuration because some systems and some faith-based schools differ.
Promotion policy	Whether progression is automatic, threshold-based, or subject-based	Threshold on session average, with school-set pass mark	Some systems promote automatically; some require passing named core subjects.
Terminal qualification	The examination the system leads towards	WAEC and NECO at senior secondary; BECE at the end of junior secondary	IGCSE, A-Level, IB Diploma, high school diploma, KCSE.
Currency and locale	Fee display, invoicing, number and date formatting	Naira, en-NG	Configuration, never a display helper hard-coded to one locale.
Dominant parent channel	Which channel notifications default to	WhatsApp	Email in the United States and much of Europe; SMS in parts

FIELD	PURPOSE	NIGERIAN VALUE	CONTRAST CASE
			of South Asia. Assuming WhatsApp globally would be wrong.

9.2.2 — Layer 2: Level Structure

A level structure is an ordered list of stages, each containing an ordered list of classes. A school selects which stages it operates; a school running nursery to senior secondary and a school running junior secondary only both use the same structure and enable different parts of it.

FIELD	MEANING
stage_code / stage_name	The grouping — for example Nursery, Primary, Junior Secondary, Senior Secondary
classes[]	Ordered classes within the stage, each with a code, display name and sequence position
is_terminal_stage	Whether the stage ends in an external examination
terminal_examination	Which examination, where applicable
stream_required_from	The class at which students divide into streams or departments, if the system does that at all
arm_supported	Whether classes subdivide into arms or sections — near-universal in Nigeria, absent in many systems
next_stage_code	Where students progress to, which is what makes end-of-session promotion computable rather than manual

ARMS ARE NOT AN AFTERTHOUGHT

A Nigerian private school of 600 students will run JSS 1A, 1B, 1C and often 1D. The arm is where a student actually sits, where attendance is marked, and what a class position is computed against. Systems that model only the class and treat the arm as a label produce class positions that are meaningless to a Nigerian parent — who wants to know where their child ranks in JSS 1B, and separately across the whole of JSS 1. Both must be computable, and which one appears on the report card is a school setting.

9.2.3 — Layer 3: Subject Framework

FIELD	MEANING
subject_catalogue[]	Every subject available in the system, with code, name, short name for broadsheets, and category
category	Core, elective, compulsory-choice (a trade or language where one must be selected from a set), or non-academic
offered_at_levels[]	Which classes a subject may be taken at
stream_restriction	Whether a subject is restricted to a stream — for example a business subject available only to commercial students
min_subjects / max_subjects	Per level, so that a school configuring a class outside the permitted range is warned at setup rather than at result computation
choice_groups[]	Sets from which a fixed number must be chosen — one Nigerian language, one trade subject, one religious study
counts_toward_average	Whether the subject contributes to the session average and position. Non-academic subjects frequently do not.

FIELD	MEANING
language_of_instruction	Where a system teaches subjects in different languages

CHOICE GROUPS ARE THE FIELD MOST SYSTEMS GET WRONG

A Nigerian junior secondary student must take exactly one Nigerian language and exactly one trade subject. A senior secondary student takes five compulsory core subjects and then a set of electives determined by their stream. Modelling all of this as a flat list of optional subjects means the platform cannot tell a school that a student is short a trade subject, and cannot compute a valid subject count. The choice group is what makes validation possible.

9.2.4 — Layer 4: Assessment Framework

This is the layer the result computation engine reads, and therefore the layer where an error becomes a wrong result on a child's report card. It is specified more tightly than the others.

FIELD	MEANING	CONSTRAINT
components[]	The named parts of a subject score — for example First Continuous Assessment, Second Continuous Assessment, Examination	Each has a code, display name, maximum raw score and weight
weight	The proportion each component contributes	Weights must total exactly 100. The system refuses to save a framework that does not, because a framework summing to 98 produces quietly wrong results for a whole term.
max_raw_score	The score out of which the component is marked	Entry above the maximum is rejected at entry, not at computation
term_aggregation	How components combine into a term score	Weighted sum by default; other rules are configuration
session_aggregation	How term scores combine into a session score	Average of terms, weighted terms, or final term only — all three are in real use
cumulative_display	Whether a term report shows a running cumulative score alongside the term score	Common and expected in Nigeria
rounding	Where and how rounding occurs	Fixed platform-wide: half up, two decimals, at final total only. This is not configurable, and it is deliberately not configurable.
position_basis	Whether class position is computed within the arm, the class, or both	Both computable; which is displayed is a report card setting
tie_handling	What happens when two students have identical totals	Equal position with the next position skipped, by default. Must be explicit — an unspecified tie rule is a guaranteed parent complaint.
absent_handling	How an absent component is treated	Distinct from a zero, always. A student absent for a test has not scored nothing; conflating the two is a misrepresentation that a parent will notice.

ROUNDING IS DEFINED ONCE, CENTRALLY, AND IS NOT A SCHOOL SETTING

Half up, two decimal places, applied at the final total only — never at each component. Two schools computing the same scores must reach the same result, and a school must reach the same result today and next term. Making rounding configurable would create a class of support ticket that is essentially impossible to resolve, because both parties would be correct.

Every computed value carries a stored calculation trace showing each component, its raw score, its weight, the intermediate figures and the final rounding step. A school forgives a slow page. It never forgives a wrong result — and the trace is how a disputed result is resolved in minutes rather than by rebuilding the calculation by hand.

9.2.5 — Layer 5: Grading Scale

FIELD	MEANING
scale_type	Percentage bands, letter grade, grade point, or descriptive band only

FIELD	MEANING
bands[]	Each with lower bound, upper bound, label, descriptor, remark text and optional grade point
pass_mark	The threshold at which a subject is passed
credit_threshold	Where a system distinguishes a pass from a credit — a distinction that decides university admission in Nigeria and therefore matters intensely to parents
applies_to_levels[]	A school may legitimately use different scales at primary and secondary level
display_precision	Whether the underlying percentage is shown alongside the band, or the band alone

- Band boundaries must be contiguous and non-overlapping. The system validates this on save; a scale with a gap between 74 and 75 would leave scores uncomputable.
- A school may create a custom scale where no template fits, but templates are offered first during setup so that most schools never have to.
- Changing a grading scale mid-session is blocked by default, and where a school insists it requires an explicit confirmation stating that previously computed results will not be retrospectively regraded. Silently regrading a term already sent to parents would be indefensible.

9.2.6 — Layer 6: Report Card Template

FIELD	MEANING
layout_code	The base design — for example standard portrait, combined primary and secondary, compact for high subject counts
sections_shown[]	Which blocks appear: subject table, component breakdown, cumulative, position, attendance summary, behavioural domains, comments, fee status
position_display	Arm position, class position, both, or none. Some systems and some schools deliberately do not rank children.
behavioural_domains[]	Affective and psychomotor traits with their own rating scale — punctuality, neatness, teamwork, and so on
comment_sources[]	Which roles supply comments — class teacher, principal, and sometimes a house master
signature_blocks[]	Who signs, and whether a digital signature image is permitted
branding	Logo and colours, where the school's tier permits it
language	The language the template renders in

POSITION DISPLAY IS A GENUINE GLOBAL FORK

Ranking children by class position is normal and expected in Nigeria, and a report card without it will be received as incomplete. In much of Europe and in many United States districts, publishing a child's rank to their parents is regarded as inappropriate and in some cases is prohibited by district policy.

This is exactly the kind of decision that must be configuration rather than a product opinion. The platform computes position always, because a school may need it internally, and displays it only where the template says so.

9.3 — The Nigerian reference template

This is the default template for the beachhead market and therefore the one that must be exactly right. It is the configuration the majority of our target schools will run on the day they sign up.

A MATERIAL CHANGE OUR V3.0 TEMPLATE DID NOT REFLECT

Nigeria's national curriculum was overhauled for the 2025/2026 academic session. Subject counts were reduced at every level, several subjects were merged or renamed, a trade subject became compulsory, and digital technologies became a compulsory subject at senior secondary. The senior secondary structure moved to five compulsory core subjects plus electives grouped into fields.

Version 3.0 of this BRD described the previous structure: Basic Science and Basic Technology as separate subjects, Civic Education, Computer Studies. A school onboarded onto that template would have found its subject list wrong on the first day of term.

This matters commercially as well as factually. A school switching systems in the middle of a curriculum transition is exactly the displacement opportunity the strategy targets, and arriving with the current structure already configured is a visible advantage over an incumbent that has not updated.

9.3.1 — Level structure

STAGE	CLASSES	TERMINAL	NOTES
Early Years	Creche, Pre-Nursery, Nursery 1–3	No	Naming varies widely between schools; the template ships a common default and expects schools to rename
Primary	Primary 1 – Primary 6	No	Lower primary (1–3) and upper primary (4–6) carry different subject sets
Junior Secondary	JSS 1 – JSS 3	Yes — Basic Education Certificate Examination	End of the basic education cycle; performance determines progression to senior secondary
Senior Secondary	SSS 1 – SSS 3	Yes — WASSCE and NECO senior school certificate	Streaming into Science, Humanities or Business begins at SSS 1

- Arms are enabled: every class may subdivide into arms, conventionally lettered
- Streaming begins at SSS 1 and is a required attribute of a senior secondary student
- Progression is threshold-based on the session average, with the pass mark set by the school

9.3.2 — Subject framework, lower and upper primary

PRIMARY 1 – 3 · MINIMUM 9, MAXIMUM 10	PRIMARY 4 – 6 · MINIMUM 11, MAXIMUM 13
English Studies	English Studies
Mathematics	Mathematics
Nigerian Language — one, from Hausa, Igbo or Yoruba (choice group)	Nigerian Language — one (choice group)
Basic Science	Basic Science and Technology
Physical and Health Education	Physical and Health Education

PRIMARY 1 – 3 · MINIMUM 9, MAXIMUM 10	PRIMARY 4 – 6 · MINIMUM 11, MAXIMUM 13
Christian Religious Studies or Islamic Studies (choice group)	Basic Digital Literacy
Nigerian History	Christian Religious Studies or Islamic Studies (choice group)
Social and Citizenship Studies	Nigerian History
Cultural and Creative Arts	Social and Citizenship Studies
Arabic Language (optional)	Cultural and Creative Arts
—	Pre-Vocational Studies, including entrepreneurial studies
—	French (optional)
—	Arabic Language (optional)

9.3.3 — Subject framework, junior secondary

JSS 1 to JSS 3: a minimum of 12 subjects and a maximum of 14.

SUBJECT	CATEGORY
English Studies	Core
Mathematics	Core
Nigerian Language — one, from Hausa, Igbo or Yoruba	Compulsory choice (one required)
Intermediate Science	Core — formed by merging the former Basic Science and Basic Technology
Physical and Health Education	Core
Digital Technologies	Core — formerly Computer Studies
Christian Religious Studies or Islamic Studies	Compulsory choice (one required)
Nigerian History	Core — reintroduced as compulsory from Primary 1 to JSS 3
Social and Citizenship Studies	Core — formed by merging the former Social Studies and Civic Education
Cultural and Creative Arts	Core
Business Studies	Core
Trade subject — one required	Compulsory choice (one required) — see 9.3.5
French	Optional
Arabic	Optional

9.3.4 — Subject framework, senior secondary

SSS 1 to SSS 3: five compulsory core subjects, plus electives drawn from the student's stream. Schools typically present 8 to 9 subjects in total.

COMPULSORY CORE — ALL STUDENTS	APPLIES
English Language	All streams
General Mathematics	All streams

COMPULSORY CORE — ALL STUDENTS	APPLIES
Citizenship and Heritage Studies	All streams — formerly Civic Education
Digital Technologies	All streams — now compulsory at senior secondary
One trade subject, carried forward from junior secondary	All streams

STREAM	ELECTIVE SUBJECTS AVAILABLE
Science	Biology, Chemistry, Physics, Agriculture, Further Mathematics, Physical Education, Health Education, Foods and Nutrition, Geography, Technical Drawing
Humanities	Nigerian History, Government, Christian Religious Studies, Islamic Studies, Nigerian Language, French, Arabic, Visual Arts, Music, Literature-in-English, Home Management, Catering Craft
Business	Accounting, Commerce, Marketing, Economics

STREAM RESTRICTION IS A REAL CONSTRAINT, NOT A SUGGESTION

Under the current structure, Economics and Marketing are restricted to commercial students. A platform that lets a science student be registered for Marketing produces a subject combination the school cannot actually present for examination, and the error surfaces at the worst possible moment — when entries are being made. Stream restriction is therefore enforced at subject assignment, with an override available to the school for genuine exceptions, recorded with a reason.

9.3.5 — Trade subjects

A trade subject is compulsory. One is chosen at junior secondary and carried forward as a core subject through senior secondary. Trade subjects for non-technical schools are the following six:

TRADE SUBJECT	TRADE SUBJECT
Solar Photovoltaic Installation and Maintenance	Beauty and Cosmetology
Fashion Design and Garment Making	Computer Hardware and GSM Repairs
Livestock Farming	Horticulture and Crop Production

- Technical colleges follow a separate and much wider set administered by the technical examinations board. The template supports this as a variant selected at school setup rather than as a different curriculum.
- The trade subject a student selects at junior secondary is stored on the student record and carried automatically into senior secondary, because re-selecting it every session is a data entry burden with no purpose

9.3.6 — Assessment framework

THERE IS NO SINGLE NATIONAL STANDARD, AND THIS IS THE FINDING THAT MATTERS MOST

Nigerian schools use materially different assessment weightings. Splits of 40:60, 30:70 and 20:80 between continuous assessment and examination are all in common use, and the choice is made by the individual school rather than prescribed nationally at secondary level.

This is precisely why the assessment framework must be configuration. A platform that ships a fixed 40:60 split will compute wrong results for a large proportion of Nigerian schools, and the schools will not discover it until report cards are already in parents' hands. The template offers a default; the school confirms or changes it during setup, and the setup wizard makes this a required step for that reason.

The template ships three ready-made frameworks. A school selects one during setup, or builds its own.

FRAMEWORK	COMPONENTS	WEIGHTS	TYPICAL USER
Balanced (40:60)	First CA, Second CA, Examination	10 / 30 / 60, or 20 / 20 / 60	The most common configuration among mid-sized private schools
Examination-weighted (30:70)	First CA, Second CA, Examination	10 / 20 / 70	Schools oriented tightly around external examination preparation
Examination-dominant (20:80)	Continuous Assessment, Examination	20 / 80	More traditional schools, and some senior secondary streams
Custom	Any components the school defines	School-defined; must total exactly 100	Schools with project work, practical marks or homework as separate components

Default configuration shipped with the Nigerian template

- Components: First Continuous Assessment (out of 20, weight 10), Second Continuous Assessment (out of 20, weight 30), Examination (out of 100, weight 60)
- Term aggregation: weighted sum to a score out of 100
- Session aggregation: average of the three terms, with a running cumulative shown on the second and third term reports
- Position: computed within arm and within class; both available, arm position displayed by default
- Tie handling: equal position, next position skipped
- Absent handling: recorded as absent, never as zero, and excluded from the component average with the report card showing the absence explicitly

9.3.7 — Grading scales

Two scales ship with the Nigerian template, because the same school routinely applies different scales at different levels.

GRADE	RANGE	DESCRIPTOR	INTERPRETATION
A1	75 – 100	Excellent	Distinction
B2	70 – 74	Very Good	Distinction
B3	65 – 69	Good	Distinction
C4	60 – 64	Credit	Credit — the threshold that matters for university admission
C5	55 – 59	Credit	Credit

GRADE	RANGE	DESCRIPTOR	INTERPRETATION
C6	50 – 54	Credit	Credit
D7	45 – 49	Pass	Pass, but below credit
E8	40 – 44	Pass	Pass, but below credit
F9	0 – 39	Fail	Fail

The senior secondary scale above mirrors the national examination bands, which is why schools want it: a parent reading a termly report in the same grades their child will receive at WASSCE can see progress in terms they already understand. The credit threshold at C6 is called out explicitly because it is the boundary Nigerian parents care about most.

GRADE	RANGE	DESCRIPTOR
A	70 – 100	Excellent
B	60 – 69	Very Good
C	50 – 59	Good
D	40 – 49	Fair
F	0 – 39	Needs improvement

The five-band scale above is the shipped default for primary and junior secondary, where the nine-band examination scale is unnecessarily granular for young children and where many schools prefer descriptive language over fine gradation.

9.3.8 — Report card template

- Standard A4 portrait layout, which fits the great majority of Nigerian private schools
- Subject table showing each component score, the term total, the grade, the class or arm average, the subject position and the teacher's remark
- Cumulative column on second and third term reports
- Attendance summary: days school opened, days present, days absent
- Behavioural domains rated on a five-point scale — punctuality, neatness, attentiveness, honesty, cooperation, and a psychomotor set covering handwriting, sports and handling of tools
- Class teacher's comment and principal's comment, drawn from a bank of suggestions the teacher may edit rather than composed from nothing thirty times over
- Next term begins on, and fee balance where the school chooses to display it
- Signature blocks for class teacher and principal

9.4 — Template versioning and the transition problem

NEW IN V4.0 — AND IT IS THE SINGLE MOST VALUABLE THING IN THIS MODULE

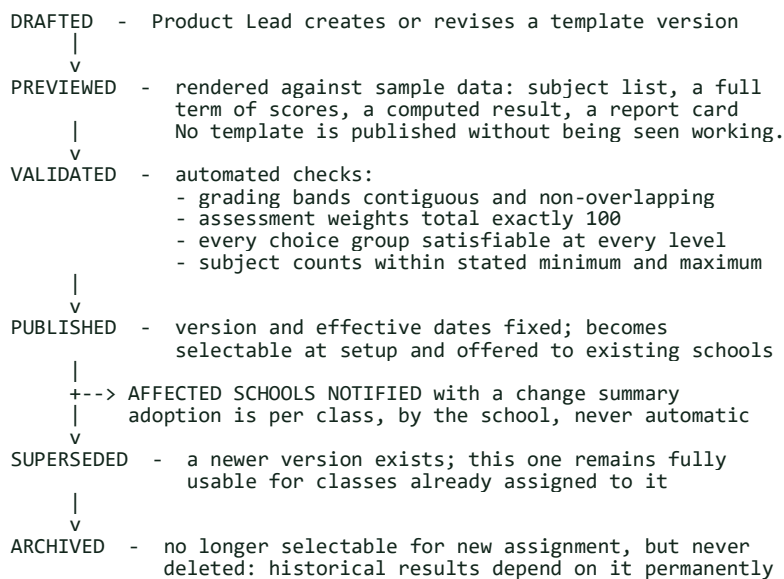
Nigeria is mid-transition. A school in the 2026/2027 session may be running the new structure at JSS 1 while its SSS 3 cohort completes under the structure they started on, because you cannot move a student into a merged subject halfway through a certificate course.

A curriculum library that holds only the current version cannot represent that school at all. It would force them to choose between misrepresenting their junior classes and misrepresenting their senior ones — and a school that

discovers this during setup will conclude, correctly, that the platform does not understand how schools actually work.

REQUIREMENT	SPECIFICATION
Templates are versioned	Every curriculum template carries a version, an effective-from date and an effective-to date. Versions are never edited in place; a change creates a new version.
Assignment is per class, not per school	A school assigns a curriculum version to a class or cohort. JSS 1 may run the 2025 structure while SSS 3 runs the 2024 structure, simultaneously, within one school.
Historical results are immutable	A result computed under one version always renders under that version. Reissuing a report card from two years ago produces the document that was originally issued, not a retrospective reinterpretation.
Migration path is explicit	Where a subject is merged or renamed between versions, the template records the mapping. A student's history shows Basic Science and Basic Technology in earlier years and Intermediate Science later, with the relationship recorded rather than the earlier records rewritten.
Schools are notified, not forced	When a new version is published, affected schools are told and shown what changed. Nothing is applied automatically. A vendor that silently changes a school's subject list has changed their results without asking.

CURRICULUM TEMPLATE LIFECYCLE



Template lifecycle — archived is not deleted, because a result computed in 2026 must still be explainable in 2032.

9.5 — Template library, current and planned

TEMPLATE	MARKET	KEY CHARACTERISTICS	STATUS
Nigerian National (NERDC 2025)	Nigeria	Three terms, September start, JSS and SSS structure, nine-band examination scale, compulsory trade subject, streaming from SSS 1	Launch — the default

TEMPLATE	MARKET	KEY CHARACTERISTICS	STATUS
Nigerian National (pre-2025)	Nigeria	The superseded structure, retained for cohorts completing under it	Launch — required for the transition
Nigerian British	Nigeria, international schools	IGCSE grading, British subject naming, three terms retained to match the local calendar	Launch
Nigerian American	Nigeria, international schools	Grade levels, grade-point scale, semester option	Launch
International Baccalaureate	Nigeria, international schools	Seven-point scale, IB subject groups, two-semester structure	Phase 2
Ghanaian National	Ghana	Three terms, Ghanaian subject list, national examination alignment	Phase 2 — a configuration record, not a build
Kenyan Competency-Based	Kenya	Competency-based assessment, descriptive bands, three terms	Phase 3
South African National	South Africa	January session start, seven-level achievement scale, four terms	Phase 3

THE KENYAN CASE IS THE REAL TEST OF THIS DESIGN

Kenya's competency-based model does not produce a percentage that averages neatly into a class position. It records achievement against competencies using descriptive bands. A platform whose result engine assumes a numeric score cannot represent it at all — not with a configuration record, not with any amount of effort short of a rewrite.

This is why Layer 5 includes a descriptive-band-only scale type, and why the result engine must treat the numeric percentage as one possible scale type rather than as the shape of the data. Building that in now costs a schema decision. Discovering it during a Kenyan launch costs the launch.

9.6 — Portability tests every template must pass

Before any template is published, and before any feature that touches curriculum, assessment or grading is merged, it is checked against these. A failure means the design is wrong, not that the template needs adjusting.

#	TEST	FAILURE LOOKS LIKE
T1	Does it work with a different currency, calendar, script or grading system?	A fee field formatted with a Naira symbol in code; a date parser assuming day-month-year
T2	Does it assume a three-term year?	A results table with three fixed columns; a session average dividing by three
T3	Does it assume examination-band grading?	A grade lookup returning A1 to F9; a report card with a column headed WAEC Grade
T4	Does it assume a numeric score exists at all?	A position computation that cannot run against descriptive bands
T5	Does it assume WhatsApp is the parent channel?	A notification path with no channel fallback; interface copy that says WhatsApp rather than the school's configured channel
T6	Does it assume a single-campus, owner-operated school?	No path for a group or trust; a single owner with no delegation model
T7	Would it survive scrutiny under a stricter regime than Nigeria's?	Children's data with no consent record; no route to erasure; no processing record
T8	Can a new country launch by adding configuration records?	Any answer that includes writing code

WHY THESE ARE IN THE BRD RATHER THAN IN AN ENGINEERING CHECKLIST

The target is not a Nigerian company that later expands. It is a globally-architected platform that begins in Nigeria because that is the market we understand best. The difference between those two things is decided in small choices made early, and the wrong ones are invisible for about three years and then require a rebuild.

Where a decision would be different for a global product than for a Nigerian one, the specification names both and the team chooses knowingly. Silent localisation is the mistake that forces the rebuild.

9.7 — Country Configuration Records

NEW IN V4.0

Several Nigeria-specific behaviours were scattered across v3.0 as prose rather than data: business registration validation, ministry approval formats, phone number validation, the blocked web address list, working hours and public holidays. Each would have become a hard-coded assumption. They are consolidated here as one record per country.

FIELD	PURPOSE	CONSUMED BY
registration_authority	Name and number format of the business registration body	M2.9 risk signals, signup form labels
education_authority	Name and approval number format of the education regulator	M2.9, signup form
phone_format	Valid patterns and recognised network prefixes	M2.9, M4 message delivery, all contact validation

FIELD	PURPOSE	CONSUMED BY
blocked_address_terms[]	Reserved words, examination and regulatory body names, competitor names, offensive terms	M2.10
default_channel	Dominant parent notification channel	M4, School Admin notification defaults
working_hours / holidays[]	Support service level calculation	M6.3
currency / locale	Formatting, invoicing, reporting	M3, M7
data_protection_regime	Which framework applies, retention defaults, consent age	M10
verification_sources[]	Which public checks a reviewer can actually perform in this country	M2.9.2 review workflow

Key Data Entities

ENTITY	KEY FIELDS
Education System Template	system_id, name, country_code, calendar_shape, term_count, session_start_month, promotion_policy, terminal_qualification, default_channel, version, effective_from, effective_to
Level Structure	structure_id, system_id, stages (json: stage_code, classes[], is_terminal, terminal_exam, stream_required_from, arm_supported, next_stage_code)
Subject Framework	framework_id, system_id, subject_catalogue (json), choice_groups (json), level_rules (json: min_subjects, max_subjects, stream_restrictions)
Assessment Framework	framework_id, system_id, components (json: code, name, max_raw_score, weight), term_aggregation, session_aggregation, cumulative_display, position_basis, tie_handling, absent_handling
Grading Scale Template	scale_id, name, scale_type, bands (json: lower, upper, label, descriptor, remark, grade_point), pass_mark, credit_threshold, applies_to_levels[]
Report Card Template	template_id, name, layout_code, sections_shown[], position_display, behavioural_domains (json), comment_sources[], signature_blocks[], language, available_to_tiers[]
Curriculum Version Assignment	school_id, class_id, cohort, template_id, template_version, assigned_at, assigned_by
Subject Mapping	from_version, to_version, from_subject_code, to_subject_code, relationship (merged/renamed/split/withdrawn)
Country Configuration	country_code, registration_authority, education_authority, phone_format, blocked_address_terms[], default_channel, working_hours, holidays[], currency, locale, data_protection_regime, verification_sources[]

Offline Behaviour

None internally. On the school side this is the most important cached data in the entire platform: a teacher marking scores offline needs the subject list, the assessment components, their maximum raw scores and the grading bands available locally. The active curriculum configuration is cached with the application on first login and refreshed on every sync. A configuration change is applied on the next successful sync and never mid-entry, so that a teacher who has entered fifteen scores under one set of maximums never has the rules change beneath them.

Notification Triggers

EVENT	RECIPIENT	CHANNEL
New curriculum version published	Every school on the affected system	Email plus in-app, with a change summary

EVENT	RECIPIENT	CHANNEL
Template validation failed	Product Lead	In-app, blocking publication
School's assessment framework does not total 100	School admin and Account Manager	In-app, blocking result computation until resolved
School attempts a grading scale change mid-session	School admin	In-app confirmation stating that prior results are not regraded

Permissions Note

- View the library: all internal roles
- Create, edit or publish a template version: Product Lead and Super Admin only
- Archive a version: Super Admin only. No role at any level may delete one, because historical results depend on it permanently.
- Edit a country configuration record: Product Lead and Super Admin
- Schools consume templates through their own setup and never touch the master library
- A school's customised copy is a separate entity owned by that school. Changes to a master template never propagate into it.

User Stories

- As an onboarding specialist, I want new schools to select from ready-made curriculum, assessment and grading templates, so that a school can be fully configured in its first session without needing support.
- As a school proprietor, I want the current national subject list already loaded when I sign up, so that I am not spending my first evening typing in fourteen subject names the platform should already know.
- As a principal, I want to run the new curriculum structure at JSS 1 while my SSS 3 cohort completes under the old one, so that the platform reflects how my school actually works during the transition rather than forcing me to misrepresent one of them.
- As a class teacher, I want the subject list, score maximums and grading bands available on my phone with no connection, so that I can mark a full class of scores in a building with no signal and have them sync later.
- As a parent, I want my child's grade shown in the same bands they will receive at their national examination, so that I can tell whether they are on track without needing it translated.
- As a product manager, I want to add a Ghanaian template ahead of a Ghana launch, so that expansion requires a configuration record rather than a development sprint.
- As a Super Admin, I want every template previewed against sample data and validated before publication, so that every option in the library is genuinely professional and computes correctly on the day a school picks it.

MODULE 10

Security, Audit & Compliance

PURPOSE

The platform's accountability layer. Every consequential action taken by anyone — Future Realm staff or school staff — is recorded permanently and cannot be altered by anybody at any level.

This module is what allows Future Realm to answer, with evidence rather than assurance, the only two questions that matter after an incident: what happened, and who did it. It is also where the platform's regulatory obligations are operated rather than merely acknowledged.

CHANGED IN V4.0 *Rebuilt around the framework that actually applies. v3.0 referenced NDPA 2023 and a general "NDPC compliance dashboard". The operative instrument changed in September 2025 and v3.0's retention claim cited a regulatory source that does not exist. Registration, DPO, DPIA, breach notification, grievance handling, cross-border records and processing records are now specified as operated processes with owners and clocks.*

10.1 — Master Audit Log

Every consequential action on the platform writes one entry. There is no action of consequence that does not.

FIELD	MEANING
Who	Actor identity, name at time of action, role at time of action, and the school membership the action occurred under
What	Action type and the entity affected
Before	Full previous state of the record
After	Full new state of the record
When	Timestamp, with time zone
Where	Address of origin, device, and approximate location
Why	Reason, mandatory for every action carrying a reason requirement

Immutability

- Enforced at the database layer, not by application logic. There is no update or delete path to an audit entry — the permission does not exist because the operation does not exist.
- No user at any access level can modify or remove an entry. This includes Super Admin, and it includes a school owner acting on their own school's records.
- Entries are retained for a minimum of 24 months, and permanently for anything touching an ownership transfer, a data correction, a support access session or a security incident.
- Written from the first commit of every module. It is not a phase and is never retrofitted — an audit trail added later is an audit trail with a hole in it exactly where the early problems are.

10.2 — Audit Search and Investigation

- Filter by actor, school, action type, entity type, date range, address of origin, and severity
- Full-text search across reasons and entity identifiers
- Timeline view reconstructing everything that happened to one record, or everything one person did in a session

- Export to Excel and CSV for formal investigation, with the export itself logged
- Saved investigations, so that a case worked across several days does not have to be reassembled each time

10.3 — Security Incident Register

SEVERITY	DEFINITION	RESPONSE
Critical	Confirmed unauthorised access to personal data, or confirmed data loss	Immediate: CTO and Super Admin paged. Breach clock starts (10.5).
High	Credible suspicion of compromise; repeated failed access on privileged accounts	Investigated within 4 hours
Medium	Policy violation without evidence of data exposure	Investigated within 24 hours
Low	Anomaly warranting a record but not urgent action	Reviewed in the weekly security review

- Every incident carries an owner, a timeline, actions taken, root cause, and a closure statement
- A feature flag rollback (M8.3) automatically opens an incident record, so that emergency changes are always accounted for afterwards
- Post-incident review is mandatory for Critical and High, with the outcome feeding the product backlog rather than a document nobody reopens

10.4 — Data Retention

CORRECTING A CLAIM MADE IN V3.0

Version 3.0 stated that school data is retained for 12 months after closure "per NDPC requirements". No such general retention period exists in the framework. Asserting a regulatory source that cannot be produced is exactly the kind of thing an audit finds, and it undermines the credibility of everything stated alongside it.

The 12-month period is Future Realm retention policy. It is defensible — it gives a school time to change its mind and to complete a migration elsewhere — but it must be stated as our policy, resting on a documented lawful basis, and it must be disclosed to the school in the data processing agreement.

DATA CLASS	RETENTION	BASIS
Active school data	For the life of the subscription	Contract
Closed school data	12 months from closure, then scheduled deletion	Future Realm policy, disclosed in the data processing agreement, to permit reactivation and migration
Accounts closed under a risk finding (M2.9)	Immediate deletion scheduling, with only the evidence of the finding retained	There is no legitimate school whose records we would be preserving
Audit entries	Minimum 24 months; permanent for the categories in 10.1	Accountability, and defence of the platform and the school
Support tickets	36 months	Service history and dispute resolution
Financial records	As required by applicable company and tax law	Legal obligation
Marketing consent and opt-out records	Retained while the record is needed to evidence that the opt-out was honoured	Compliance evidence

- Deletion is scheduled, executed and then evidenced. A deletion that cannot be evidenced has not, for practical purposes, occurred.
- A school is notified before its retention clock expires, with a final opportunity to export

10.5 — Regulatory Operations

THE FRAMEWORK THAT ACTUALLY APPLIES, AND WHAT IT MEANS FOR US

Nigeria's General Application and Implementation Directive was issued on 20 March 2025 and took effect on 19 September 2025. On its issuance the previous 2019 regulation and its 2020 implementation framework ceased to apply. The Data Protection Act 2023 and the Directive together are now the complete framework.

Registration as a controller or processor of major importance is triggered at more than 200 data subjects in a six-month window. The tiers are Ordinary High Level (200–999), Extra-High Level (1,000 and above) and Ultra-High Level (5,000 and above). Our client schools sit at the ordinary level. Future Realm, aggregating students, guardians and staff across every school on the platform, reaches the highest tier almost immediately — the tier carrying the largest registration fees and an annual compliance audit filing that the ordinary tier does not face.

Separately, the Directive defines a child as anyone under 18 and requires parent or guardian consent to process a child's data regardless of the purpose of processing. The Onboarding Specification's under-13 consent trigger is the United States standard and is wrong here. Consent is universal for school students, not conditional on age.

The practical conclusion: registration and the appointment of a Data Protection Officer are launch requirements, not later commitments, and their cost belongs in the financial model now.

10.5.1 — Obligations register

OBLIGATION	CADENCE	OWNER	TRACKED IN
Registration as a controller and processor of major importance	At launch, then per the applicable renewal or filing cycle for our tier	Data Protection Officer	M1.7 countdown
Annual compliance audit filing	Annual, per our tier	Data Protection Officer with Finance	M1.7 countdown
Data Protection Officer appointed and details published and registered	At launch; on any change within the notification window	Super Admin	10.5
Data protection impact assessment	Before launch, and again before any materially new processing	Data Protection Officer with CTO	10.5.2
Record of processing activities	Maintained continuously	Data Protection Officer	10.5.3
Breach notification	Within the applicable window from becoming aware	CTO and Data Protection Officer	10.5.4
Grievance mechanism	Continuous; every grievance logged and answered	Support Lead with Data Protection Officer	10.5.5
Cross-border transfer record	Maintained per destination and reviewed on any change of provider or region	CTO with Data Protection Officer	10.5.6
Data processing agreement with every school	At onboarding; re-accepted on version change and on ownership transfer	Legal with Onboarding	M2.3 compliance tab

10.5.2 — Data Protection Impact Assessment

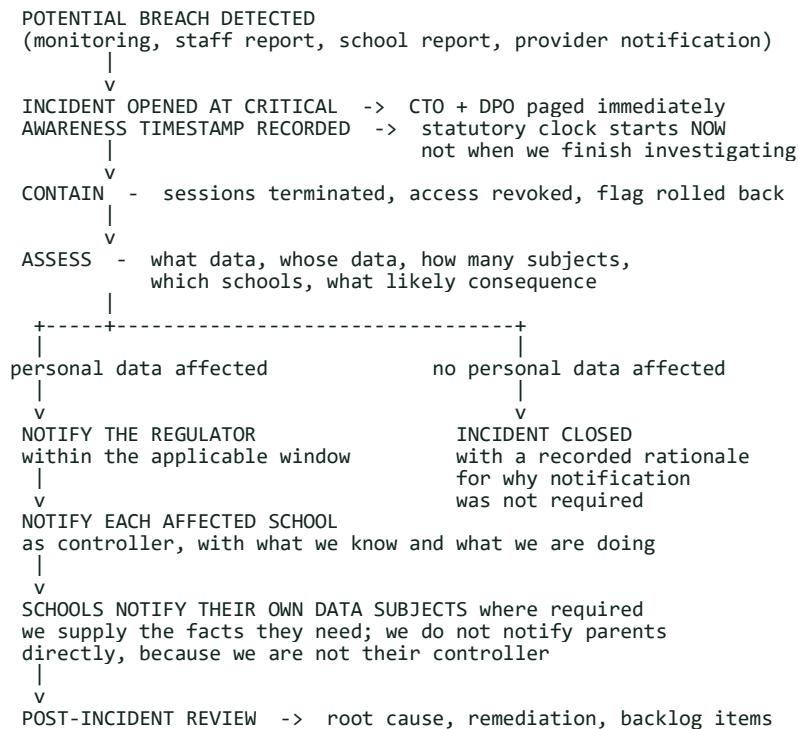
- Required before launch. The platform processes children's data at scale, across many schools, including sensitive categories such as medical and disciplinary records — this is high-risk processing by any reading.
- Held as a living record in this module, not a document filed once and forgotten
- Reassessed before any materially new processing: a new module handling sensitive data, a new country, a change of hosting region, or any feature placing Future Realm in the controller position
- The free teacher tier is explicitly named as requiring its own assessment before it ships, for the reason recorded in Open Issue 2

10.5.3 — Record of Processing Activities

- What personal data the platform holds, for whom, on what basis, for how long, and who it is shared with
- Maintained per processing activity rather than as one narrative document, so it can be produced on request without being rewritten
- Automatically cross-referenced to the data entity tables in each module, so that a new entity holding personal data cannot be added without appearing here

10.5.4 — Breach notification

BREACH HANDLING — THE CLOCK STARTS AT AWARENESS, NOT AT CONFIRMATION



Breach handling — note the direction of notification: to the school, which is the controller, not around it.

10.5.5 — Grievance mechanism

- A published route by which any data subject — a parent, a member of staff, a former student — can raise a concern about how their data has been handled
- Every grievance logged with a reference, an owner and a response clock; open items appear on M1.7 and cannot be dismissed
- Where the grievance concerns a school's own processing, it is routed to that school as controller with our assistance, and the routing is recorded
- Outcomes reviewed monthly for patterns — the same complaint from three schools is a product defect, not three complaints

10.5.6 — Cross-border transfer record

- Where data is hosted, processed, backed up and supported from, per region
- Reviewed on any change of hosting region, provider or support arrangement, and disclosed to schools in the data processing agreement
- Relevant now rather than later: notification and email providers, backup destinations and support tooling frequently sit outside the country the school is in, and a school asked to name where its children's data lives must be able to get an accurate answer from us

10.6 — Data Subject Request Handling

REQUEST TYPE	ROUTE	OUR ROLE
Access — what data is held on me or my child	Through the school, as controller	We assist the school in producing it; the school responds
Correction	Through the school	M6.6 data correction process
Erasure	Through the school	Assessed against retention obligations; never executed unilaterally at school level and never executed by us without the controller's instruction
Objection or withdrawal of consent	Through the school for school data; directly to us for our own marketing	Immediate for marketing; assisted for school data
Portability	Through the school	Satisfied by the school's own full export in Excel and CSV

- Every request is logged with its statutory clock, and appears on M1.7 with days remaining
- A request arriving directly at Future Realm for school-held data is acknowledged, routed to the school within one working day, and tracked to completion — never left with the requester to chase

10.7 — Compliance Dashboard

- Registration and filing status with next deadline
- Impact assessment status and date of last review
- Open data subject requests and open grievances with their clocks
- Data processing agreement coverage: schools on the current version, schools on an outdated version, schools not yet accepted
- Consent coverage across the platform, by school — a school with low guardian consent coverage is a risk to both of us
- Open incidents and unresolved breach obligations, shown permanently and never dismissible

Key Data Entities

ENTITY	KEY FIELDS
Audit Entry	entry_id, school_id, actor_id, actor_name, actor_role, membership_id, action_type, entity_type, entity_id, before_state (json), after_state (json), reason, ip_address, device, location_approx, timestamp
Security Incident	incident_id, severity, detected_at, awareness_at, detected_by, owner_id, timeline (json), root_cause, remediation, closed_at, notification_required (bool)
Retention Schedule	data_class, school_id (nullable), retention_period, basis, clock_started_at, deletion_due, deletion_executed_at, evidence_ref
Regulatory Obligation	obligation_id, name, cadence, owner_role, last_completed_at, next_due_at, evidence_ref, status
Impact Assessment	assessment_id, scope, risks (json), mitigations (json), residual_risk, assessed_by, assessed_at, next_review_due

ENTITY	KEY FIELDS
Processing Activity	activity_id, purpose, data_categories[], subject_categories[], lawful_basis, recipients[], retention_ref, transfers_outside_country (bool)
Breach Record	breach_id, incident_id, awareness_at, regulator_notified_at, schools_notified_at, subjects_affected_count, data_categories[], outcome
Grievance	grievance_id, raised_by, school_id (nullable), subject, raised_at, owner_id, response_due, responded_at, outcome
Data Subject Request	request_id, request_type, requester, school_id, received_at, routed_at, statutory_due, completed_at, outcome
Cross-Border Record	region, data_categories[], destination_country, provider, safeguard_mechanism, reviewed_at

Offline Behaviour

None internally. On the school side the requirement is stricter than elsewhere: an action performed offline must carry its audit entry in the sync queue alongside it, so that the entry arrives with the change rather than being generated at sync time. An audit entry written at sync time records when we found out, not when it happened — which is precisely the wrong information.

Notification Triggers

EVENT	RECIPIENT	CHANNEL
Critical incident opened	CTO, Super Admin, Data Protection Officer	Immediate page plus email
Breach notification obligation live	CTO, DPO, Super Admin	Persistent dashboard item plus email until discharged
Affected school notified of a breach	School Proprietor and Principal	Email plus phone call from a named person
Regulatory deadline at 30, 14 and 7 days	Data Protection Officer	Email
Data subject request at 7 days remaining	Data Protection Officer and Support Lead	In-app plus email
Retention clock expiring for a closed school	School's last known contact	Email, final export opportunity
Audit export performed	Super Admin	In-app, always — exporting the audit log is itself audited

Permissions Note

- View full audit log across all schools: Super Admin, CTO and Data Protection Officer
- View audit log for own schools: Account Manager, Portfolio scope
- Export audit records: Super Admin only, and the export is itself audited
- Modify or delete an audit entry: nobody. The function does not exist at any level.
- Manage retention schedules and regulatory obligations: Data Protection Officer and Super Admin
- Open and manage security incidents: CTO, Support Lead and above
- The Data Protection Officer holds Global scope over this module and read access elsewhere, with no billing rights at all. Independence is the point of the role.

User Stories

- As a Super Admin, I want every consequential action recorded with before and after values, so that any dispute is settled by evidence rather than by recollection.
- As a CTO, I want audit entries to be impossible to alter, including by me, so that the log is worth something precisely because nobody can tidy it.

- As a data protection officer, I want every regulatory deadline counted down on the dashboard, so that a statutory clock is never discovered after it has run out.
- As a school principal, I want to know exactly where my students' data is stored and who has access to it, so that I can answer a parent's question honestly and immediately.
- As a parent, I want a route to raise a concern about my child's data that is actually answered, so that the school and its vendor are accountable to me and not only to each other.
- As a founder, I want a defensible compliance position rather than an intention, so that regulatory exposure never becomes the reason a good product fails.

MODULE 11

Internal Team & Role Management

PURPOSE

Controls who inside Future Realm can access the admin panel and what each person can do. As the team grows from three founders to a department structure, precise internal access control becomes as important as school-facing security — and considerably easier to get wrong, because everyone internally is trusted.

It is the module that operates the access control foundation specified in Section A.

CHANGED IN V4.0 *Extended from two dimensions to three. Every internal account now carries a data scope alongside module and function access. Onboarding and offboarding made explicit processes rather than assumed ones.*

11.1 — Internal User Registry

- Every Future Realm staff account with role, department, data scope, status, last login and account creation date
- Actions: create, edit, suspend, reactivate, delete, force password reset, end all sessions
- Filter by department, role, scope and status

11.2 — Permission Configuration

A grid combining all three dimensions. Each cell is set independently, per person.

MODULE	ACCESS LEVEL	FUNCTION RIGHTS	DATA SCOPE
Per module, individually	None / View / Edit / Full	Named functions within the module, each toggled separately	Global / Region / Portfolio / Assigned / Partner / Own / None

- Changing a permission takes effect on the person's next request, not their next login. A revoked permission that persists until logout is not revoked.
- A "what this person can see" preview is shown before any permission set is saved, rendering the panel exactly as that person would experience it. This is what prevents accidental over-permissioning, and it is the same control the school-facing portal uses when adding staff.
- Every change is written to M10 with before and after states
- No role may grant a permission or a scope it does not itself hold. Self-elevation is blocked at the data layer.

11.3 — Role Templates

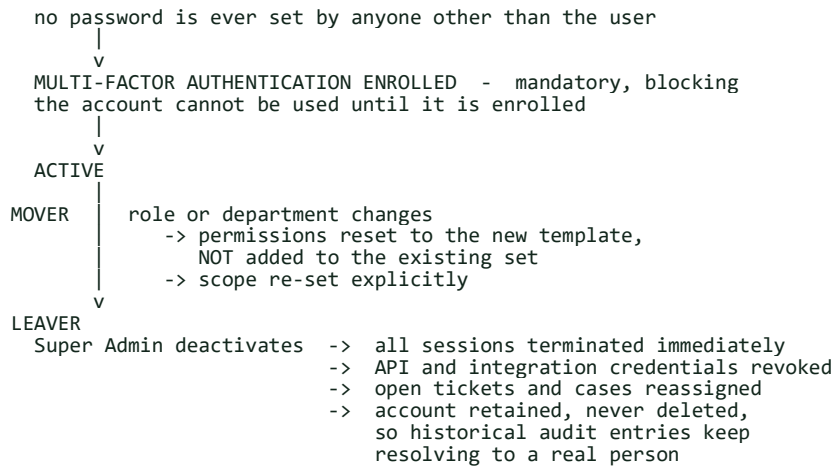
- Pre-built sets for the standard roles, so that a new team member is configured in one action rather than twenty
- Custom roles can be created as the team grows — a Regional Growth Lead is defined by narrowing scope, not by writing code
- Changing a template prompts, but never forces, application to people already holding it. Silently widening someone's access because a template changed is an access control failure.

11.4 — Onboarding and Offboarding**INTERNAL ACCOUNT LIFECYCLE****JOINER**

Super Admin creates the account, assigns role template, sets data scope explicitly (never inherited silently)



INVITATION SENT - single use, 24 hours, elevated role



Internal lifecycle — the mover case is where access quietly accumulates in most organisations.

WHY A LEAVER'S ACCOUNT IS DEACTIVATED AND NEVER DELETED

Deleting the account would leave two years of audit entries attributed to an identifier that resolves to nothing. The audit log's value depends on every actor remaining identifiable for as long as the entries are retained. Deactivation removes all access while preserving accountability.

11.5 — Internal Activity Monitoring

- Every internal action logged to M10 with no exceptions and no suppression option
- Per-person activity view: modules accessed, actions taken, schools touched, support access sessions run
- Anomaly flags: access outside normal working hours, unusually large exports, access to schools outside a person's normal pattern, repeated permission escalation requests
- Weekly summary to Super Admin covering permission changes, new accounts, and anomaly flags

11.6 — Internal Security Requirements

REQUIREMENT	SPECIFICATION
Multi-factor authentication	Mandatory for every internal account without exception. Enrolment blocks first use.
Session timeout	30 minutes of inactivity, no exceptions and no per-user override
Panel address	A separate, non-public address, not linked from any public page
Address restriction	Configurable allow-list, Super Admin only, for additional restriction
Password policy	Minimum 12 characters for internal accounts, last five blocked, no forced periodic rotation
Credential sharing	Prohibited and detectable — concurrent sessions from distant locations raise an anomaly flag
Device registration	Known devices recorded; a login from an unrecognised device requires a second factor and notifies the user

Key Data Entities

ENTITY	KEY FIELDS
Internal User	user_id, name, email, department, role_template_id, data_scope, scope_values[], status, mfa_enrolled_at, created_by, created_at, deactivated_at

ENTITY	KEY FIELDS
Permission Set	set_id, user_id, module_access (json), function_rights (json), data_scope, scope_values[], granted_by, granted_at
Role Template	template_id, name, department, default_module_access, default_function_rights, default_scope, ceiling_level
Internal Activity Record	record_id, user_id, module, action, school_id (nullable), timestamp, anomaly_flag

Offline Behaviour

None. Permission state is evaluated server-side on every request and is never cached on an internal device.

Notification Triggers

EVENT	RECIPIENT	CHANNEL
Internal account created	New user and Super Admin	Email
Permission or scope changed	The affected user and Super Admin	Email plus in-app
Anomaly flag raised	Super Admin and CTO	In-app plus email
Account deactivated	Super Admin and the user's Department Lead	Email
Weekly access summary	Super Admin	Email

Permissions Note

- Create, edit or delete internal accounts: Super Admin only
- Adjust permissions within own department, up to own ceiling: Department Lead
- View internal activity for own department: Department Lead
- View all internal activity: Super Admin and CTO
- Grant a scope wider than one's own: not permitted for any role

User Stories

- As a Super Admin, I want to grant precise access per person across all three dimensions, so that every team member has exactly what their work requires and nothing more.
- As a Super Admin, I want a preview of what a person will actually see before I save their permissions, so that nobody is over-permissioned by accident.
- As a department lead, I want to adjust my own team's access within my ceiling, so that I can manage my team without waiting on a founder.
- As a CTO, I want every internal action logged with no way to suppress it, so that internal accountability matches the standard we hold schools to.
- As a regional lead, I want my scope set once so every screen is already narrowed to my region, so that I work with my own business rather than filtering a national view daily.
- As a Super Admin, I want a leaver's account deactivated rather than deleted, so that two years of audit history still resolves to a real person.

MODULE 12

System & Infrastructure Monitoring

PURPOSE

Technical health monitoring for the CTO and engineering. It exists so that Future Realm knows about a problem before any school does — which, in a market where trust is the scarcest asset, is the difference between an incident and a reputation.

It is also where the offline-first architecture is proved to be working rather than assumed to be.

CHANGED IN V4.0 *Provider names removed from thresholds and moved to per-region configuration. Result computation health added as a first-class monitored concern, because it is the one failure the product cannot survive.*

12.1 — System Health Dashboard

- Server response time, average and 95th percentile
- Database query performance and slow query log
- Error rate by module
- Active user sessions across the platform
- Storage utilisation and growth trend, with a projected exhaustion date rather than a bare percentage
- Uptime, current month and rolling twelve months

12.2 — Offline Sync Monitoring

The most important monitoring on the platform, because offline-first is the primary technical differentiator and a silent sync failure is indistinguishable from data loss for the school experiencing it.

- Total records currently pending sync across all schools
- Average age of the oldest queued record
- Sync failure rate over 24 hours
- Failure reasons ranked, so the team fixes the largest cause first
- Per-school sync status, showing which schools have records pending
- Conflict rate — records that required conflict resolution rather than a clean apply

METRIC	WARNING	CRITICAL
Platform-wide sync failure rate	Above 5% of attempts in one hour	Above 15% of attempts in one hour
Single school sync failures	3 consecutive failures	10 or more consecutive failures
Oldest unsynced record	More than 2 hours	More than 6 hours
Conflict rate	Above 1% of synced records	Above 5% of synced records

Operational reading: if average queue age exceeds two hours, a group of schools has been offline for an extended period and may need proactive contact. If the failure rate spikes, it indicates a server-side problem requiring immediate investigation rather than school-by-school support.

12.3 — Result Computation Health

NEW IN V4.0

The platform's own principles state that result correctness is existential — a school forgives a slow page and never forgives a wrong result. Version 3.0 monitored servers, sync and messaging, and did not monitor the one thing that cannot be allowed to fail.

- Golden-dataset regression suite executed on every deployment; a failure blocks the release outright and cannot be overridden
- Computation volume and duration by school, so that a school computing an entire term's results is not silently timing out
- Trace completeness: the proportion of computed values carrying a full calculation trace. Anything below 100% is a critical alert, because a value without a trace cannot be defended to a parent.
- Recomputation events, with reason — a school recomputing published results repeatedly is a signal of either a configuration problem or a data problem, and either way somebody should call them
- Assessment frameworks failing validation across the platform, which surfaces schools whose weights no longer total 100 before they compute anything

12.4 — Notification Delivery Monitoring

Infrastructure-level monitoring, distinct from Module 4's campaign tracking. This module confirms the connections are working; Module 4 confirms the messages are landing. Together they separate "the provider is down" from "the message did not resonate".

CHANNEL	WARNING	CRITICAL
Email delivery failure rate	Above 3%	Above 10%
SMS delivery failure rate	Above 3%	Above 10%
WhatsApp delivery failure rate	Above 5%	Above 15%

12.5 — Backup and Disaster Recovery

BACKUP TYPE	FREQUENCY	RETENTION	VERIFICATION
Full database backup	Daily	30 days	Automatically verified after each run
Incremental backup	Every 6 hours	7 days	Automatically verified
Configuration backup	On every configuration change	Last 10 versions	Manually verified monthly
Offsite copy	Daily	90 days	Manually verified monthly

- A manual backup can be triggered at any time
- The backup log shows the last 30 events with status and file size
- A missed or failed backup fires a critical alert immediately and cannot be cleared until a successful backup completes
- Restore is tested quarterly against a real restore into an isolated environment. A backup that has never been restored is a hypothesis, not a backup.
- The disaster recovery procedure is linked from this module and reviewed quarterly

12.6 — Third-Party Integration Status

Providers are named in the per-region configuration record (M9.7), never in this specification. The module monitors whichever provider is configured for the region.

INTEGRATION	CHECK FREQUENCY	ON FAILURE
WhatsApp business messaging provider	Every 5 minutes	Alert CTO and pause all WhatsApp sends
SMS gateway (per region)	Every 5 minutes	Alert CTO and fall back to email
Email service (per region)	Every 5 minutes	Alert CTO
Payment provider (Phase 2)	Every 5 minutes	Alert Finance Lead and CTO
Hosting and storage region	Every minute	Alert CTO, page on sustained failure

- A paused channel is surfaced to schools in the platform interface rather than failing silently, because a school that believes it has notified parents and has not is worse off than one that knows it must wait

12.7 — Feature Flag Status

- Quick view of every active flag, its stage, and its rollout percentage
- Links to M8 for management
- Any flag in pilot or partial state beyond 30 days is flagged for review

Key Data Entities

ENTITY	KEY FIELDS
System Health Metric	metric_name, value, threshold_warning, threshold_critical, status, timestamp
Sync Failure Log	school_id, module, failure_reason, occurred_at, retry_count, resolved (bool), resolved_at
Sync Conflict Record	school_id, entity_type, entity_id, resolution_rule_applied, resolved_at
Computation Health Record	school_id, computation_type, records_computed, duration_ms, trace_complete (bool), recomputation_of (nullable), reason
Delivery Health Metric	channel, region, provider_ref, success_rate, failure_count, failure_reasons[], period
Backup Record	backup_id, type, status, file_size, completed_at, verified (bool), storage_location, restore_tested_at
Integration Status	integration_name, region, last_successful_call, current_status, last_failure_reason, paused (bool)

Offline Behaviour

None internally. The entire purpose of this module is to observe the school-side offline architecture from a position that is itself always online.

Notification Triggers

EVENT	RECIPIENT	CHANNEL
Any critical threshold breached	CTO	Immediate page plus email
Backup missed or failed	CTO and Super Admin	Immediate, persistent until resolved
Regression suite failure on deployment	CTO and Product Lead	Immediate; the release is blocked
Trace completeness below 100%	CTO	Immediate — critical
Integration paused	CTO, Support Lead, and affected schools in-app	Immediate
Daily infrastructure summary	CTO	Email, 07:00

Permissions Note

- Full module access: CTO and Super Admin only
- Simplified health status on the dashboard: all internal roles
- Trigger a manual backup: CTO and Super Admin only
- View per-school sync failure logs: Support Lead and above, to diagnose school-reported issues
- View computation health for own schools: Account Manager, Portfolio scope

User Stories

- As a CTO, I want to be alerted the moment sync failure rates spike across several schools, so that I investigate before dozens of schools separately report the same issue.
- As a CTO, I want daily confirmation that backups completed and quarterly proof that they restore, so that a worst-case data loss scenario is recoverable in fact rather than in theory.
- As a CTO, I want the result computation regression suite to block a release on failure, so that a wrong result never reaches a child's report card.
- As an operations lead, I want delivery failures separated from campaign performance, so that I can tell the difference between a provider outage and a message nobody opened.
- As a support lead, I want to see which schools have records stuck in the sync queue, so that we contact them proactively rather than waiting for them to notice a discrepancy.

MODULE 13 · NEW

Partner, Channel & Commission Management

PURPOSE

Distribution is roughly half of this business. The product does not sell itself in this market, partnerships with education consultancies and associations are a primary channel, and partner economics must be modelled with the same rigour as product architecture.

This module exists because the counter-proposal already issued to our first distribution partner commits Future Realm to deal registration, attribution rules, clawback provisions, commission on collected and reconciled revenue only, and a termly statement of account — and none of it existed anywhere in the specification. Every one of those commitments would have been operated on a spreadsheet, which is how partner disputes begin.

CHANGED IN V4.0 Entire module new in v4.0.

13.1 — The economics this module encodes

RECURRING SOFTWARE AND ONE-OFF PROJECTS ARE DIFFERENT BUSINESSES

A completed website costs nothing after delivery. A live school costs infrastructure, storage, support, messaging, maintenance and compliance every day it remains on the platform. Any partner proposing a single blanket rate across both has not understood the economics, and correcting that understanding is the first move in any negotiation.

The system must therefore support at least two structurally different commission models simultaneously, and must never allow one to be applied to the other.

	STREAM A — AGENCY AND IT SERVICES	STREAM B — SCHOOL MANAGEMENT SUBSCRIPTIONS
Nature	Project-based, delivered once	Recurring subscription, per academic term
Cost to us after delivery	Effectively zero	Substantial and permanent
Commission shape	Flat share of project value	Declining rate across a defined term, then ends
Basis	Collected and reconciled project revenue	Collected and reconciled subscription revenue, fully paid invoices only
Settlement	Monthly	Termly, within an agreed window after term close

Three rules the system enforces rather than trusts

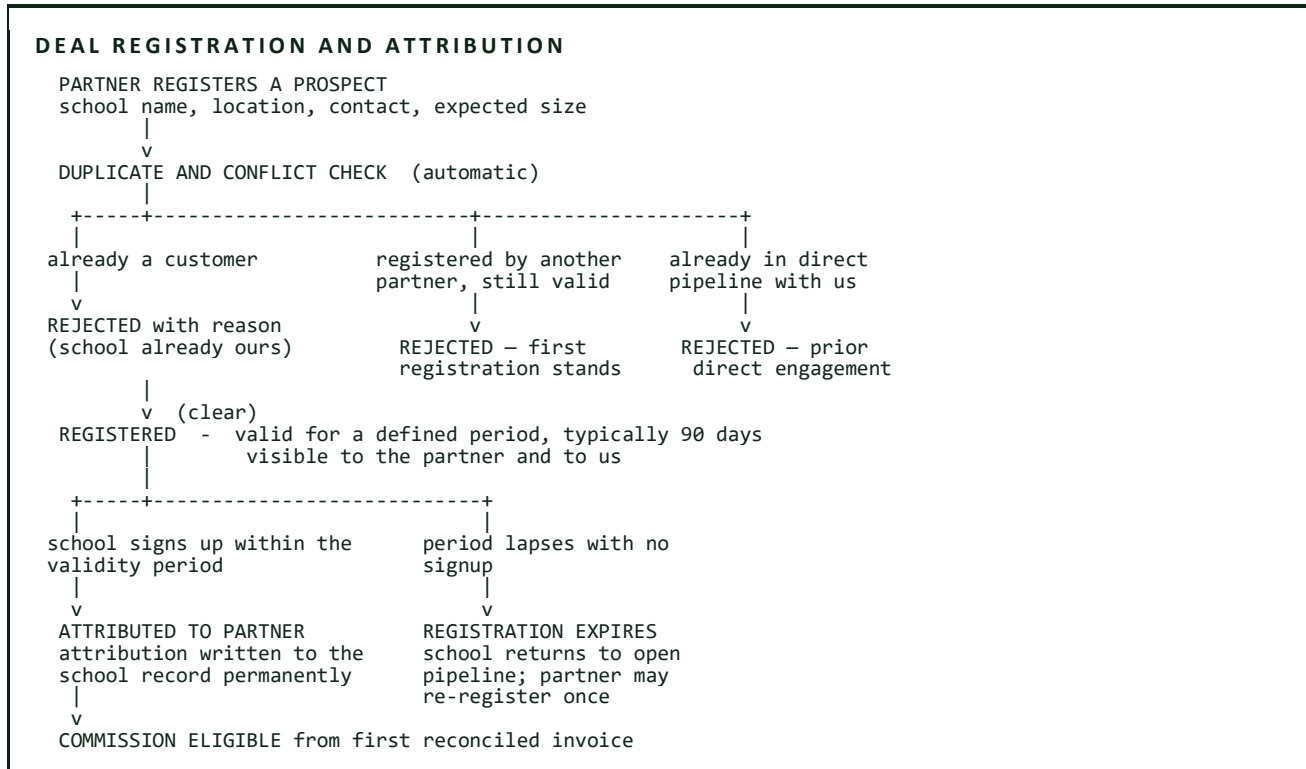
- Commission accrues on reconciled revenue only. Not on invoiced amounts — an invoice is a statement of what is owed, not income received.
- Commission accrues on fully settled invoices only. A partial payment accrues nothing until the balance is settled.
- A commission rate belongs to a stream. The system will not apply an agency rate to a subscription invoice, or the reverse, because that is the single error most likely to occur and least likely to be noticed.

13.2 — Partner Registry

- Every partner with legal name, trading name, territory, contacts, agreement version, term start and end, current status
- Agreement terms held as structured data, not as an uploaded document alone: commission rates by stream and year, settlement cadence, deal registration validity period, clawback window, non-compete scope, minimum performance condition
- The signed agreement is attached, and the structured terms are what the system actually operates on. A rate that lives only inside a PDF is a rate somebody will eventually type in wrongly.
- Status: Prospective, Active, Under Review, Notice Period, Concluded

13.3 — Deal Registration

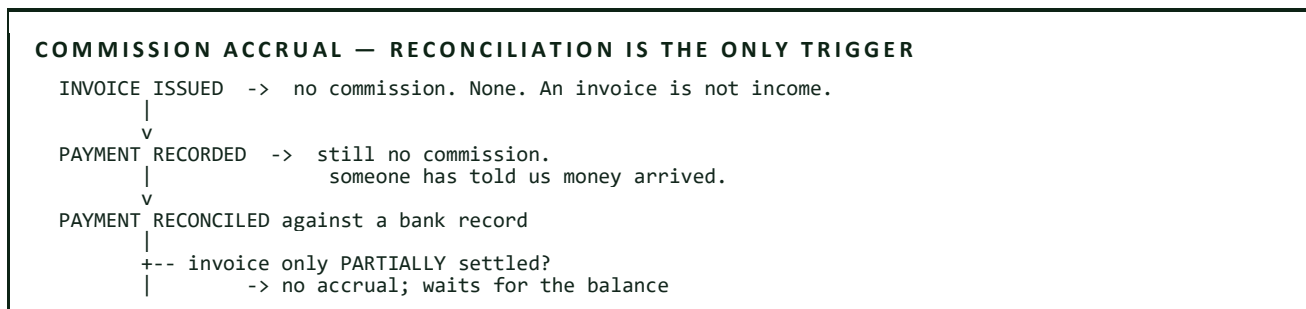
Attribution is established before the introduction, never argued about afterwards. This is the mechanism that prevents the most common and most damaging partner dispute: two parties claiming the same school.

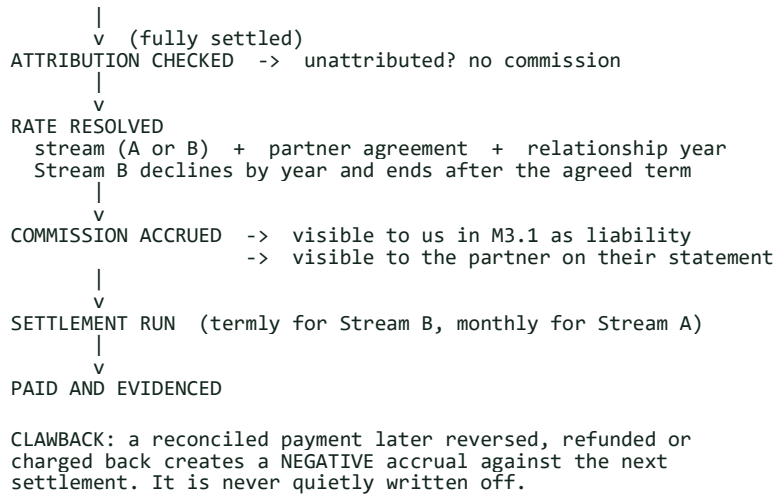


Deal registration — first valid registration wins, and the rule is visible to every party in advance.

- Attribution is written to the school record at sign up and does not change afterwards except by a documented attribution dispute resolution
- A school arriving through self-signup with no registration is unattributed, and remains so. A partner cannot claim a school retrospectively.
- The Individual Users Register (M5.9) routes leads to partners only under a registration, so that the introduction and the attribution happen in the same action

13.4 — Commission Calculation





Commission accrual — the whole design exists to make reconciliation, not invoicing, the trigger.

RULE	SPECIFICATION
Basis	Collected and reconciled revenue only
Partial payments	No accrual until the invoice is fully settled
Rate by stream	Stream A flat; Stream B declining by relationship year and ending after the agreed term
Relationship year	Counted from the school's first reconciled invoice, not from the agreement date — so a school signed in month eleven still carries a full first-year rate
Clawback	Reversal, refund or chargeback creates a negative accrual against the next settlement
Post-termination	Commission continues on schools already onboarded, for their agreed cycle, conditional on the partner continuing to meet its support obligations
Discounts	Commission is computed on the discounted amount actually collected, never on list price
Credits and add-ons	Notification credit purchases are treated as subscription revenue for commission purposes only where the agreement says so — the default is that they are not

13.5 — Partner Statement of Account

- Produced each settlement period, showing every attributed school, invoices reconciled in the period, amounts collected, rate applied, commission accrued, clawbacks, and net payable
- Every line traceable to an invoice and a reconciliation record. A statement a partner cannot verify is a statement they will dispute.
- Issued whether or not the amount is positive, so that a zero period is a statement rather than a silence
- Historic statements retained permanently and reproducible exactly as issued

13.6 — Partner Performance

- Schools registered, converted and retained, by partner
- Conversion rate from registration to signup, and from signup to second term
- Retention of partner-sourced schools against direct-sourced schools — the number that reveals whether a channel is producing customers or merely signatures
- Support ticket volume from partner-sourced schools, which tests whether first-line support obligations are actually being met
- Minimum performance condition tracked automatically: where no new school is registered across two consecutive terms, both parties are alerted that the review condition in the agreement has been reached

13.7 — Attribution Disputes

- Raised by a partner or by us, with the disputed school and the grounds
- Resolved against the registration record, which is timestamped and immutable — this is why registration exists
- Outcome recorded permanently against the school and the partner, with reasons
- Where a dispute is upheld, commission is recalculated from the point of attribution, forwards and backwards, and the correction appears on the next statement rather than being netted off silently

Key Data Entities

ENTITY	KEY FIELDS
Partner	partner_id, legal_name, trading_name, territory[], contacts[], status, agreement_version, term_start, term_end, non_compete_scope, notice_period
Commission Agreement	agreement_id, partner_id, stream, rate_schedule (json: year, rate), basis, settlement_cadence, registration_validity_days, clawback_window_days, post_termination_terms
Deal Registration	registration_id, partner_id, school_name, location, contact, registered_at, expires_at, status, rejection_reason, converted_school_id
Attribution	school_id, partner_id, registration_id, attributed_at, stream, relationship_year_start, dispute_ref (nullable)
Commission Accrual	accrual_id, partner_id, school_id, invoice_id, reconciliation_id, amount_collected, rate_applied, commission_amount, accrued_at, settlement_id (nullable), is_clawback (bool)
Partner Statement	statement_id, partner_id, period, lines[], gross_commission, clawbacks, net_payable, issued_at, settled_at, payment_reference
Attribution Dispute	dispute_id, school_id, raised_by, grounds, evidence_refs[], outcome, decided_by, decided_at, recalculation_ref

Offline Behaviour

None. Commission is computed server-side from reconciled financial records and can have no other authority.

Notification Triggers

EVENT	RECIPIENT	CHANNEL
Deal registration accepted or rejected	Registering partner	Email, with the reason where rejected
Registration expiring in 14 days	Registering partner and Sales Lead	Email
Attributed school converts to paid	Partner and Sales Lead	Email
Statement of account issued	Partner	Email with the statement attached
Clawback applied	Partner and Finance Lead	Email, itemised
Minimum performance condition reached	Partner and Super Admin	Email

Permissions Note

- View partner registry and performance: Sales Lead and above
- Create or amend a partner agreement: Super Admin only
- Register or reject a deal: Sales and Onboarding Leads and above
- View commission amounts: Finance Lead, Sales Lead and Super Admin
- Issue or settle a statement: Finance Lead and Super Admin
- Resolve an attribution dispute: Super Admin only

- Partner-scoped internal users see one partner's data only — the foundation of the self-serve partner portal deferred to Phase 2

User Stories

- As a finance lead, I want commission to accrue only when a payment is reconciled against a bank record, so that we never pay a partner a share of money we have not received.
- As a sales lead, I want deal registration to reject a school another partner already registered, so that attribution is settled before an introduction rather than argued about after a sale.
- As a partner, I want a termly statement itemising every attributed school and every invoice, so that I can verify what I am owed instead of trusting a single number.
- As a Super Admin, I want separate commission structures for project work and for subscriptions, so that we never pay a recurring-revenue rate on a one-off delivery, or the reverse.
- As a founder, I want retention of partner-sourced schools compared with direct-sourced ones, so that I can tell a real distribution channel from an expensive introduction service.
- As a finance lead, I want a refunded payment to create a negative accrual automatically, so that clawback is a system behaviour rather than an awkward conversation.

MODULE 14 · NEW

Data Migration & Onboarding Operations

PURPOSE

This is a displacement market. Roughly half of the genuinely addressable segment already owns something, and the opportunity is schools that bought a system, found it painful, and are stuck. Free, effortless data migration is therefore a sales weapon, not a feature.

Version 3.0 contained no migration capability of any kind — no pipeline, no quality gate, no reconciliation, no tooling. The central mechanism of the market strategy had no software behind it. This module supplies it, and takes on the operational side of onboarding that the Onboarding Specification assumes exists.

CHANGED IN V4.0 Entire module new in v4.0.

14.1 — Why migration is treated as a sales function

THE OBJECTION THIS MODULE IS DESIGNED TO ANSWER

A proprietor unhappy with their current system does not stay because they like it. They stay because moving means re-entering eight hundred students, three years of results and a fee ledger by hand, at the same time as running a school. The switching cost, not the incumbent product, is what holds them.

Which means the migration experience is not a support process that happens after the sale. It is the argument that wins the sale. "We will move everything for you, free, and show you a reconciliation proving nothing was lost" is a stronger close than any feature comparison — and it is only credible if we can actually do it repeatably.

14.2 — Migration Job Registry

- Every migration with school, source system, scope, assigned specialist, status, records expected, records imported, records rejected, and reconciliation outcome
- Filter by source system, status, specialist and age
- A blocked job older than 48 hours appears on M1.10 and contributes to the school's churn risk score — a displaced school whose data never arrived has not actually joined

14.3 — Source System Adapters

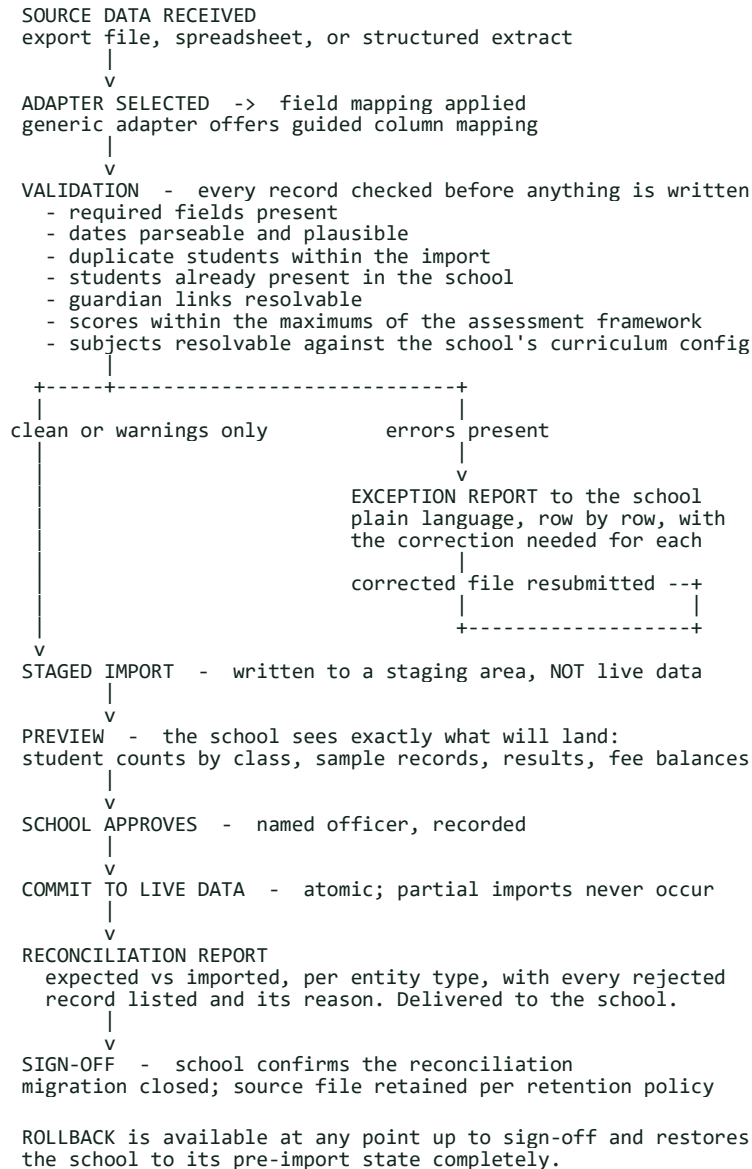
Each adapter is a configuration record describing how to read one source, not bespoke code per school.

FIELD	PURPOSE
source_system	The system being migrated from, or Spreadsheet, or Paper
accepted_formats[]	Export formats the adapter reads — spreadsheet layouts, delimited files, structured exports
field_mappings	Source field to platform field, with transformations
known_quirks	Documented behaviours of that source — date formats, how it encodes a withdrawn student, whether it exports arms separately
validation_profile	Which validation rules apply, and which warnings are expected rather than exceptional for this source
completeness_expectations	Which entities that source typically cannot export, so a specialist knows in advance what will need collecting another way

- Adapter coverage is a competitive metric, tracked in M7.7. A competitor whose exports we cannot read is a competitor we cannot displace, whatever our product does.
- A generic spreadsheet adapter with a guided column-mapping interface handles everything else, because a large share of target schools are migrating from spreadsheets rather than from a system

14.4 — Migration pipeline

MIGRATION — NOTHING ENTERS LIVE DATA WITHOUT A RECONCILIATION THE SCHOOL SIGNS OFF



Migration pipeline — staging, preview, approval, reconciliation, sign-off. No step is optional.

THE RECONCILIATION REPORT IS THE PRODUCT

The technically difficult part of migration is not moving records. It is proving to a proprietor that nothing was lost — because they have almost certainly been burned before, and because they are handing over the only copy of their students' academic history.

A reconciliation report that says "812 students expected, 812 imported, 0 rejected; 2,436 result records expected, 2,431 imported, 5 rejected and listed below with reasons" is what converts anxiety into confidence. It should be treated as a customer-facing deliverable and designed accordingly, not as a log file.

14.5 — Migration Data Handling

- Source files are personal data about children from the moment they arrive. They are stored encrypted, access-restricted to the assigned specialist and their lead, and never held on a staff device or in a personal mail folder.
- Retention: source files are deleted 90 days after sign-off, with the reconciliation report retained. The report is the evidence; the raw file is a liability once its purpose is served.
- Every access to a source file is logged to M10
- A data processing agreement must be in place with the school before any source file is accepted. There is no informal migration "just to take a look".
- Where a school has not yet converted to paid, migration data is subject to the same protections as live data. Trial status does not lower the standard.

14.6 — Sample Data Provisioning

LOAD-BEARING, NOT DECORATIVE

A 30-day trial ends long before a school's real payoff moment, which arrives at term end when results are computed and report cards go home. Sample data is what moves that moment from month three to hour one, and it is the reason the 30-day trial is viable at all. It must not be descoped if the build runs late.

LOADED CONTENTS	DETAIL
Students	30 sample students with full profiles and guardians
Staff	5 sample teachers with subject assignments
Structure	One class with arms, a full subject set, a complete timetable
Attendance	A full term already marked
Scores	Complete assessment component scores across all subjects
Results	Computed results with calculation traces, class positions and subject averages
Report cards	30 finished report cards ready to preview, download or print
Broadsheet	A complete class broadsheet ready to export
Fees	A fee structure with payments recorded and outstanding balances
Notifications	Sample result alerts, fee reminders and receipts ready to send

Controls

- Every sample record is clearly labelled as sample data wherever it appears
- Excluded from all real reports, analytics and platform metrics
- Removable in one action, and removed automatically once the school enrolls 20 real students
- **Synthetic provenance is mandatory.** Sample names, photographs and guardian details are wholly synthetic and are never derived from any real school. Sample guardian phone numbers and email addresses are non-routable, so that a trial admin testing a result notification cannot reach a real person. This was unstated previously and is the kind of omission that produces an incident nobody intended.
- Sample data is generated against the school's own selected curriculum configuration, so a British-curriculum school sees a British-curriculum report card rather than a Nigerian one

14.7 — Setup Progress Operations

- Every school's guided setup progress: which of the required steps are complete, which remain, and how long since the last step
- Incomplete setup after 5 days alerts the assigned Account Manager, as the Onboarding Specification requires
- Aggregate view showing which setup step schools most often abandon at — this is the highest-value onboarding metric available and it tells the product team exactly where the wizard is failing
- Bulk outreach to schools stalled at the same step, using a template from M4

14.8 — Invitation Operations

The back-office controls the Onboarding Specification requires for accounts Future Realm creates after a closed deal.

- Resend an invitation — a new link is issued and every previous link dies immediately
- Correct an email address and resend, with the reason recorded
- Mark as delivered manually, where delivery was confirmed by another means
- Cancel a pending invitation
- Never-activated register: every invited account that has not activated, by school and by role, with age. This is the register the Onboarding Specification points at, and it must also exist on the school side in the School Admin Portal — see Section 0.1.
- Resend limits are enforced by the system: five per account per 24 hours, matching the specification

Key Data Entities

ENTITY	KEY FIELDS
Migration Job	job_id, school_id, source_system, adapter_id, scope[], assigned_to, status, records_expected (json), records_imported (json), records_rejected (json), started_at, committed_at, signed_off_at
Source Adapter	adapter_id, source_system, accepted_formats[], field_mappings (json), known_quirks, validation_profile, completeness_expectations
Source File	file_id, job_id, filename, encrypted_ref, uploaded_at, uploaded_by, access_log_ref, delete_due, deleted_at
Validation Exception	exception_id, job_id, row_ref, entity_type, severity, rule_failed, message, resolved (bool)
Reconciliation Report	report_id, job_id, entity_counts (json), rejected_records (json), generated_at, delivered_at, signed_off_by, signed_off_at
Sample Data Set	set_id, school_id, curriculum_template_id, loaded_at, cleared_at, auto_cleared (bool)
Setup Progress	school_id, steps_complete[], steps_remaining[], last_step_at, alert_sent_at, abandoned_at_step
Invitation Record	invitation_id, user_id, school_id, role, issued_at, expires_at, resend_count, corrected_email_from, cancelled_at, activated_at

Offline Behaviour

None internally. On the school side, a completed migration must be fully available offline on the next sync — a school that has just moved its entire history onto the platform and cannot open it in a building with poor connectivity has not experienced the migration as successful.

Notification Triggers

EVENT	RECIPIENT	CHANNEL
Migration job started	School admin and assigned specialist	Email
Validation exceptions require school action	School admin	Email with the exception report attached

EVENT	RECIPIENT	CHANNEL
Preview ready for approval	School admin	Email plus in-app
Migration committed	School admin	Email with the reconciliation report
Job blocked more than 48 hours	Migration Specialist and Onboarding Lead	In-app plus email
Sample data auto-cleared at 20 real students	School admin	In-app
Setup incomplete after 5 days	Assigned Account Manager	In-app plus email
Source file scheduled for deletion	Assigned specialist	Email, 7 days before

Permissions Note

- View migration registry: Onboarding and Support roles and above, scope-narrowed
- Run a migration job: Migration Specialist and above, on assigned jobs only
- Access a source file: the assigned specialist and their lead only, always logged
- Commit to live data: requires the school's recorded approval — no internal role may commit without it
- Roll back a migration: Onboarding Lead and above, up to sign-off
- Create or edit a source adapter: CTO and Product Lead
- Invitation operations: Onboarding and Support roles and above

User Stories

- As a proprietor stuck on a painful system, I want Future Realm to move all my students, results and fee records for me, free, so that the cost of switching stops being the reason I stay somewhere I dislike.
- As a proprietor, I want a reconciliation report proving exactly what arrived and what did not, so that I can trust that three years of my students' academic history survived the move.
- As a migration specialist, I want validation to run before anything touches live data, so that a bad source file never becomes a corrupted school.
- As a school administrator, I want to preview and approve the import before it commits, so that nothing enters my school's records without my say-so.
- As a sales lead, I want adapter coverage tracked by source system, so that I know which competitors we can displace easily and which we currently cannot.
- As an onboarding lead, I want to see which setup step schools most often abandon at, so that we fix the wizard where it actually fails rather than where we assume it does.
- As a data protection officer, I want source files encrypted, access-logged and deleted 90 days after sign-off, so that a spreadsheet of eight hundred children does not sit indefinitely in a staff mailbox.

SECTION B

Non-Functional Requirements

B.1 — Performance

- Every admin panel page loads within 2 seconds under normal server load
- Data tables above 500 rows use pagination or virtual scrolling — no admin action is delayed by rendering a large dataset
- Live-data widgets use a push mechanism rather than page refresh
- Report generation above 10 seconds runs as a background job with a notification on completion, rather than holding a browser tab

B.2 — Security

- The panel is reachable only at a separate, non-public address, not linked from any public page
- Multi-factor authentication is mandatory for every internal account, with no exemption at any level
- Sessions expire after 30 minutes of inactivity, with no per-user override
- All data in transit encrypted to a current standard; all sensitive fields encrypted at rest
- Configurable address allow-list for additional restriction, Super Admin only
- Permission checks are performed server-side on every request. A hidden interface element is never treated as a control.

B.3 — Audit and accountability

- Every action by every internal user is logged to Module 10 automatically, with no suppression option
- Audit entries are immutable at the database layer — no user at any level can modify or delete them
- Minimum 24-month retention, and permanent retention for ownership transfers, data corrections, support access sessions and security incidents

B.4 — Availability

- 99.5% monthly uptime target for the admin panel
- Planned maintenance communicated to the team at least 24 hours in advance
- The admin panel remains available during school-facing maintenance windows wherever technically feasible, because that is exactly when the team needs to see what is happening

B.5 — Scalability

- Every table and report must perform within its stated threshold at 10,000 schools on the platform, not merely at current scale
- No architectural decision in this panel may create a bottleneck requiring a rebuild as the school base grows
- Data scope predicates must be indexed, since they are applied to every query in the system

B.6 — Portability

- No module may contain a hard-coded reference to a country, currency, curriculum, calendar, grading scale, provider or language
- Every such value is read from a configuration record (M9)
- The portability tests in Section 9.6 apply to every feature touching curriculum, assessment, grading, notification or identity

Section C · Out of Scope for Version 1.0

Confirmed future enhancements, explicitly excluded to protect build timeline and quality. Excluded means excluded — each of these returns as a scoped decision, not as scope creep.

EXCLUDED FROM V1.0	RETURNS WHEN
Automated payment reconciliation against a payment provider	Phase 2, when school-facing payment integration is live. Until then, reconciliation is the manual dual-control process in M3.4.
Predictive churn modelling	Phase 2 analytics. The behavioural scoring in M3.6 is deliberately rule-based and explainable, because a churn score nobody can explain is a score nobody acts on.
Automated invoice generation and delivery without admin review	After the billing module has been validated across at least two full term cycles
Self-serve partner portal	Phase 2. M13 already carries Partner data scope, which is the foundation it will be built on.
Mobile version of the admin panel	Not planned. This is a desktop back-office tool and building a mobile version would widen our exposure for no operational gain.
Automated regulatory reporting to national examination or education authorities	Phase 3, as a government partnership. The schema is designed to be exportable and mappable now; no integration is built before there is a relationship to justify it.
Multi-branch and school group management	Phase 2. The nullable group identifier is carried on the school record from the first commit.
Third-party commercial campaigns to parents	Only in the redesigned form in M4.8, in which the school opts in as controller and shares revenue. Not at all in the form v3.0 proposed.

Closing note

This specification is written to be built from directly, depended on daily by schools, and read by an investor. It assumes it will be reopened in three years and must still hold up.

Where a decision could go two ways, this version states both and chooses one with a reason attached, so that the reasoning survives the people who made it. Where something is unresolved, it is in the Open Issues Register rather than smoothed over — because the register is short and honest, and a specification that pretends otherwise is more dangerous than one that admits what it does not yet know.

The standard being aimed at is not the best school management system in Nigeria. That is the first milestone. The standard is the reliability of enterprise software, the interface quality of consumer products, and the data discipline of financial systems — applied to a platform holding the academic records of children, in schools that cannot afford for it to be wrong.

DOCUMENT	VERSION	STATUS
Platform Admin Panel (Back Admin) — Master BRD	4.0 · August 2026	Approved for build, subject to the Open Issues Register